

EE104 Fundamentals of Electric Circuits (Lesson 01)

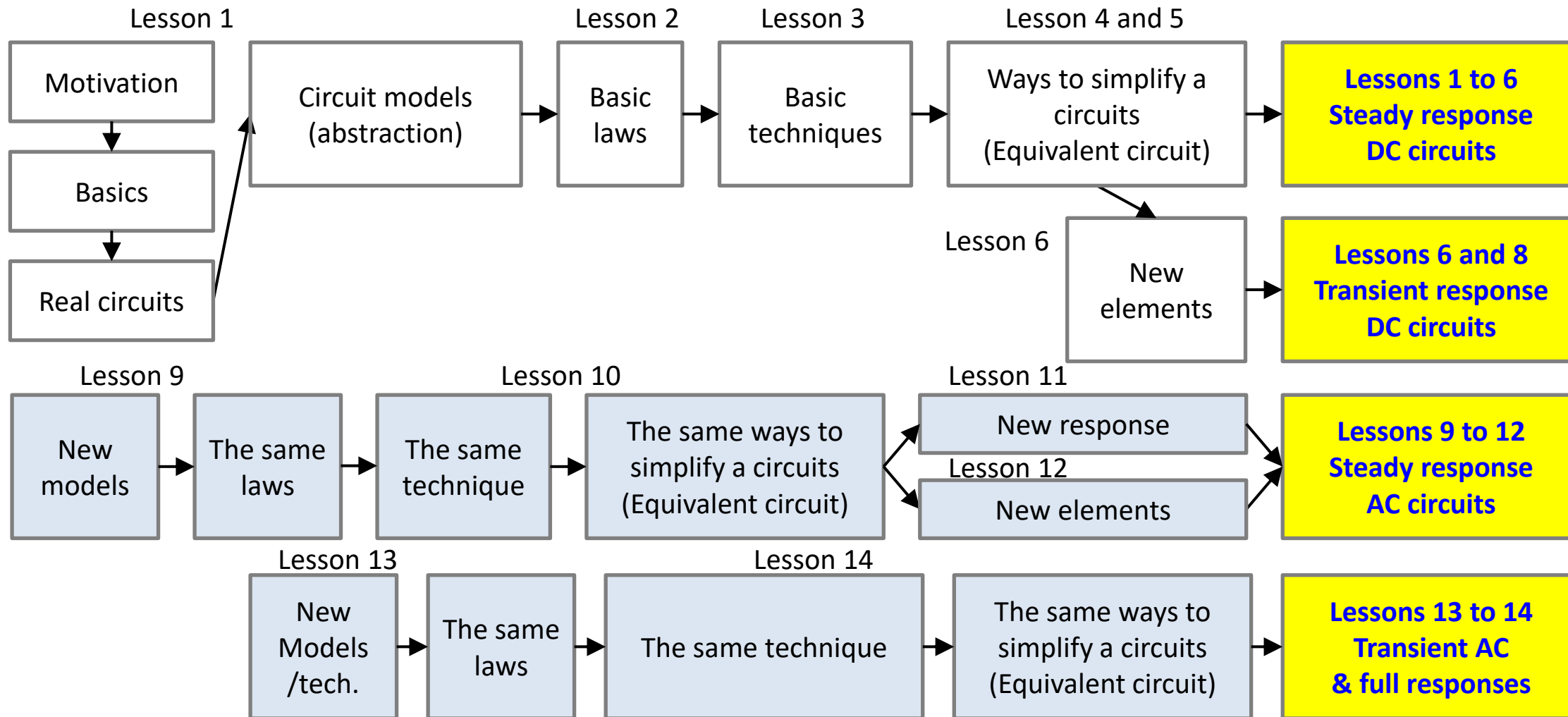
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南方科技大学-工学院-电子与电气工程系-PN Lab

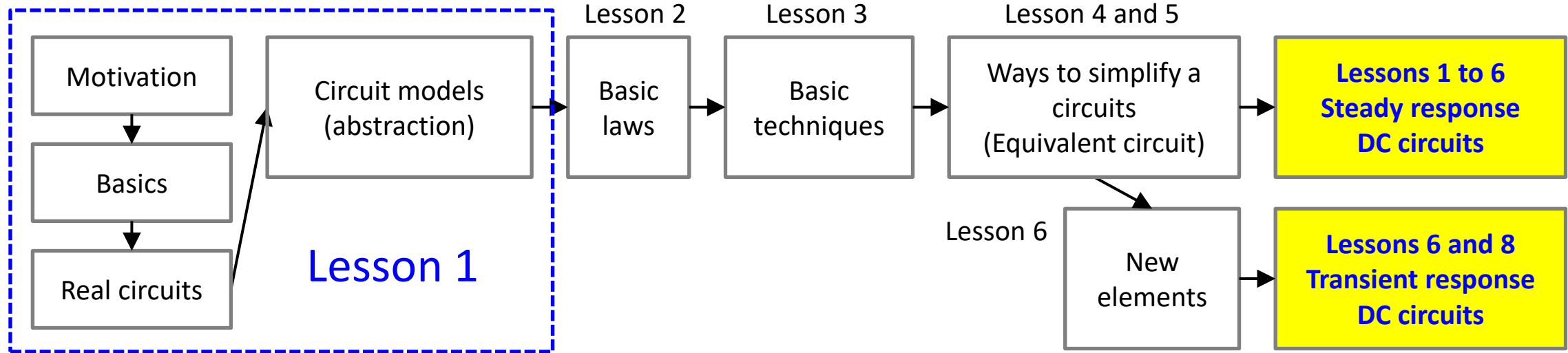
What you will learn in this course

- EE104 roadmap



Lesson 01 the basics

- EE104 roadmap (DC portion)



- a) **Electric circuits:** a circuit is an interconnection of electrical elements, which is necessary for almost every application that requires electricity to work.
- b) **Basic quantities:** current (A), voltage (V), power (W). To analyze a circuit is to find the current through its elements and the voltage across these elements.
- c) **Basic elements:** resistors, electric wires, dependent sources, independent sources.
- d) **Circuit schematics and models:** circuits based on lumped circuit abstraction.

Chapter 1 Basic Concepts

1.1-1.2	Electricity and electric circuits	Slides 05-12
1.3-1.5	Electric currents, voltage and power	Slides 14-37
1.6	Systems of Units	Slides 39-42
1.7-1.9	Circuit elements	Slides 44-56
1.10-1.11	Circuit schematics and models	Slides 58-70
1.12	Summary	Slides 72-73

1.1 Electricity (a)

- What is electricity ?



Merriam-Webster dictionary: Electricity is a fundamental form of **energy**

新华字典：可通过化学的或物理的方法获得的一种**能**

1.1 Electricity (b)

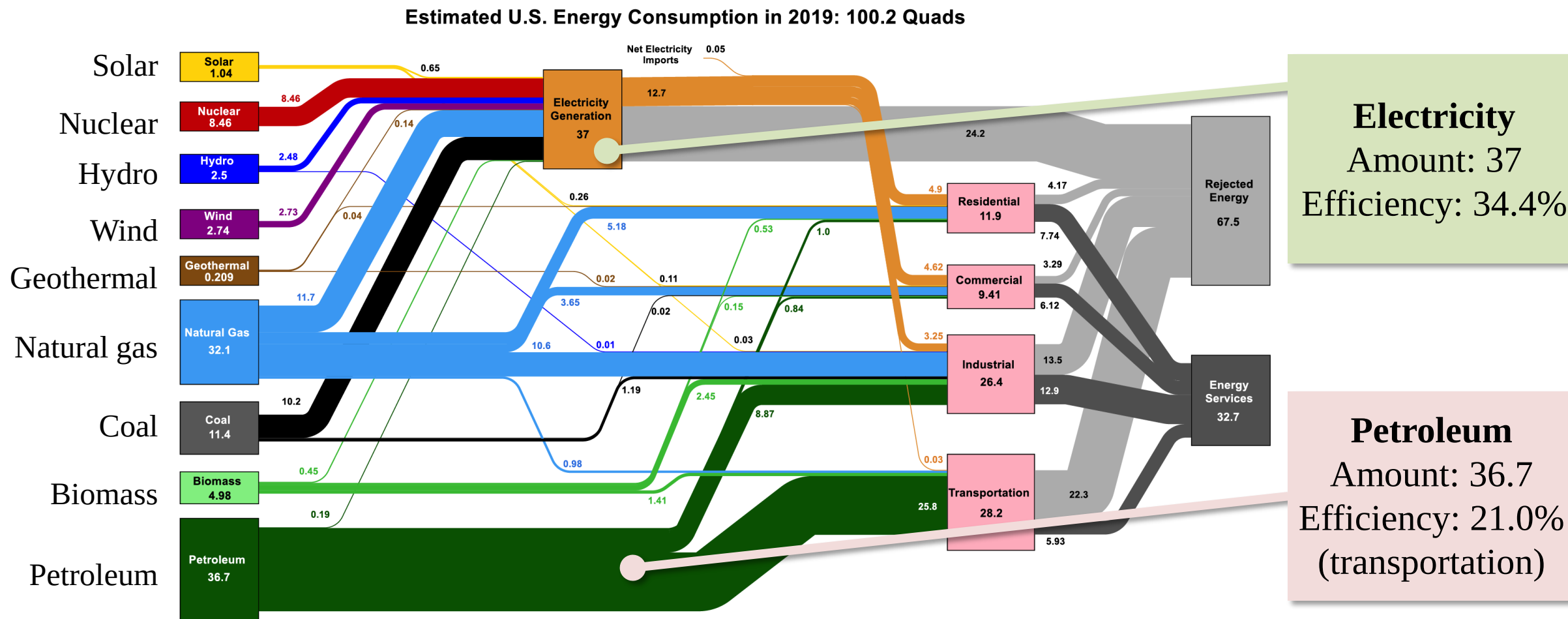
- Other energies in addition to electricity



- Advantages of electricity over other energies
 - ✓ Easy to transport and distribute
 - ✓ Highly adaptable
 - ✓ Easy transmission and process of information (by semiconductors)
 - ✓ Efficient and eco-friendly

1.1 Electricity (c)

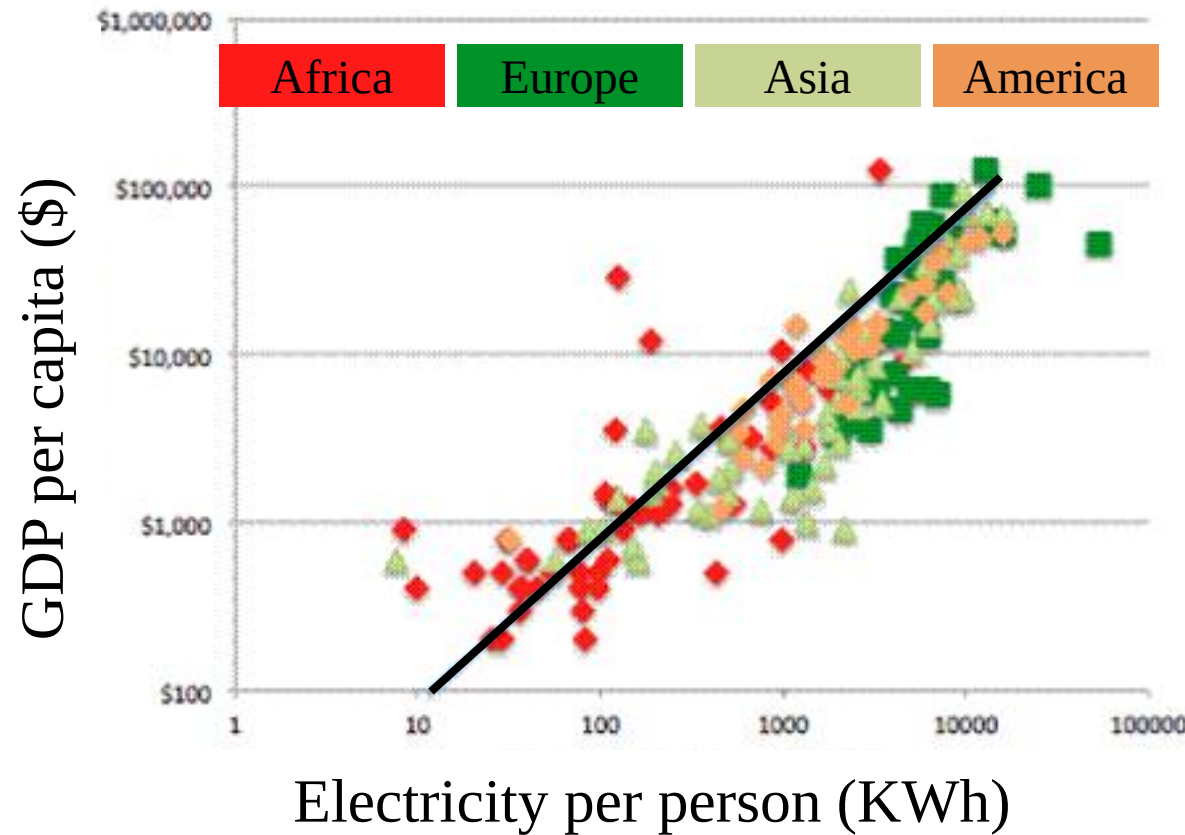
- Ultra-High importance of electricity



1.1 Electricity (d)

- GDP per capita is proportional to the consumption of electricity

GDP versus electricity consumption



1.2 Electric Circuits (a)

- How to use electricity?

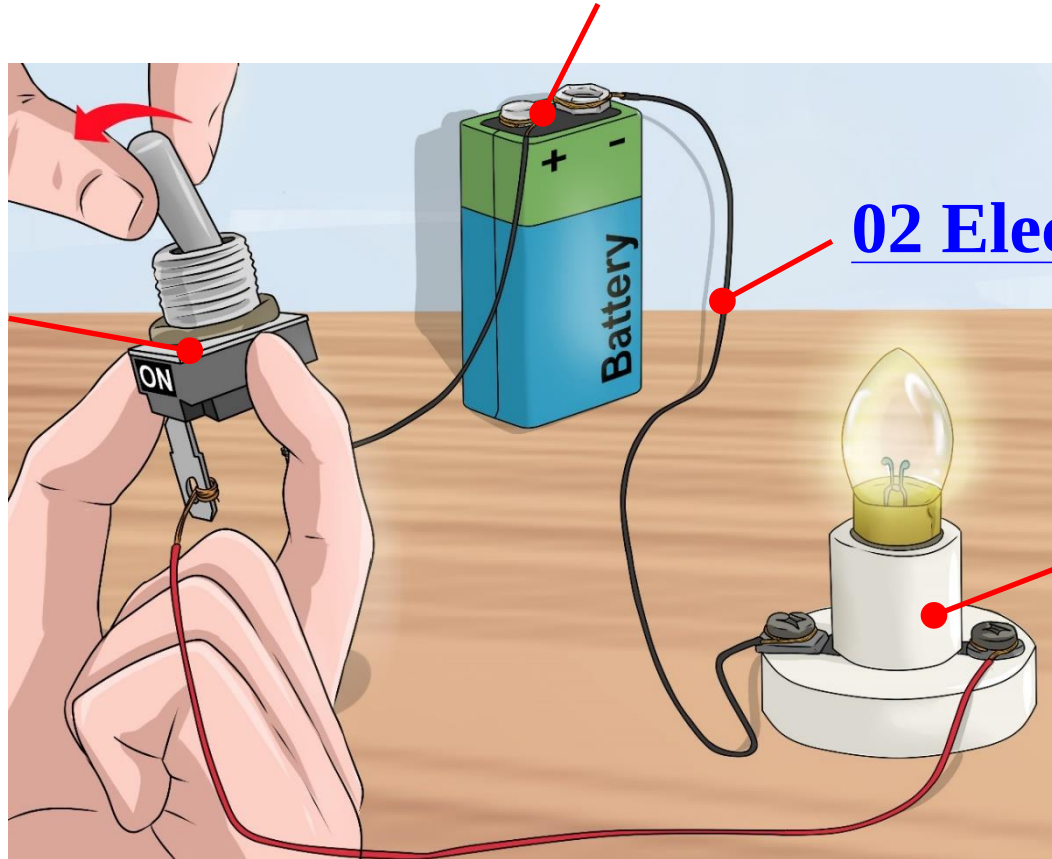
01 Electric source: providing electricity

04 Electric control:

Control or conversion
of electricity

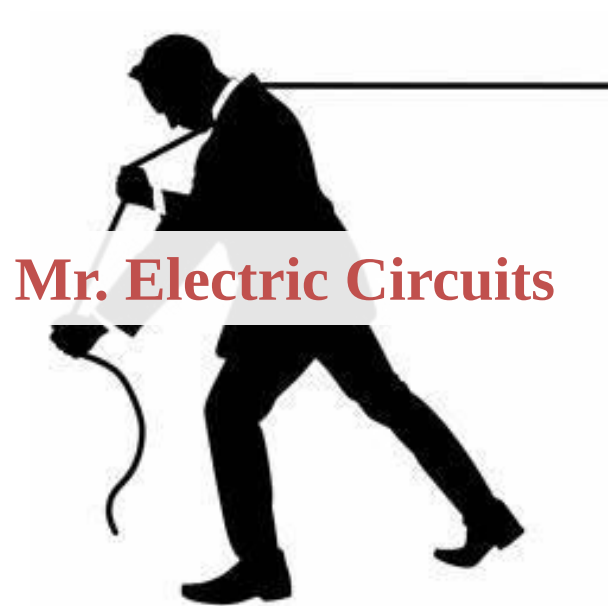
02 Electric path: flow of energy

03 Load: energy conversion
or signal processing



1.2 Electric Circuits (b)

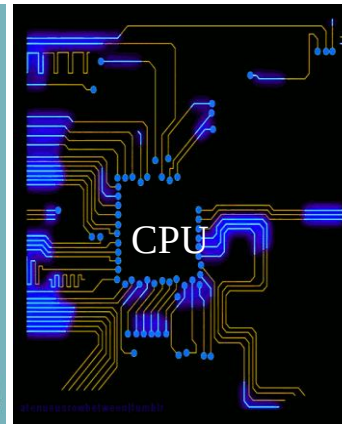
- What is an electric circuit?
 - a) An electric circuit (电路) is an interconnection of electrical elements.
 - b) Generally, the function of an electric circuit is either to communicate information (signal) or transfer power (energy) from one point to another.



Communication



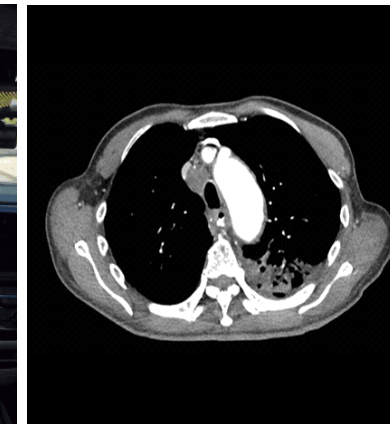
Computing



Control



Signal-processing



Electric power



1.2 Electric Circuits (c)

- Electric circuits at different scales

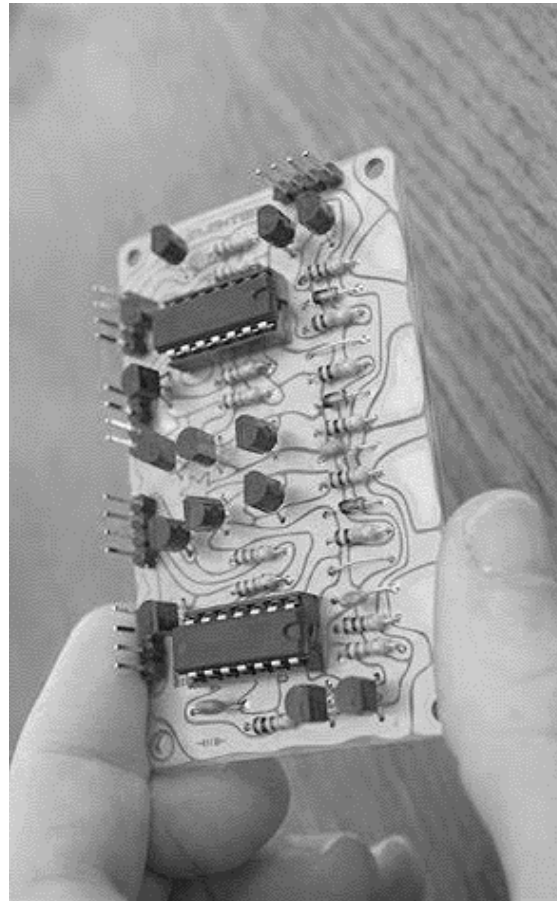
01 HV transmission ~km



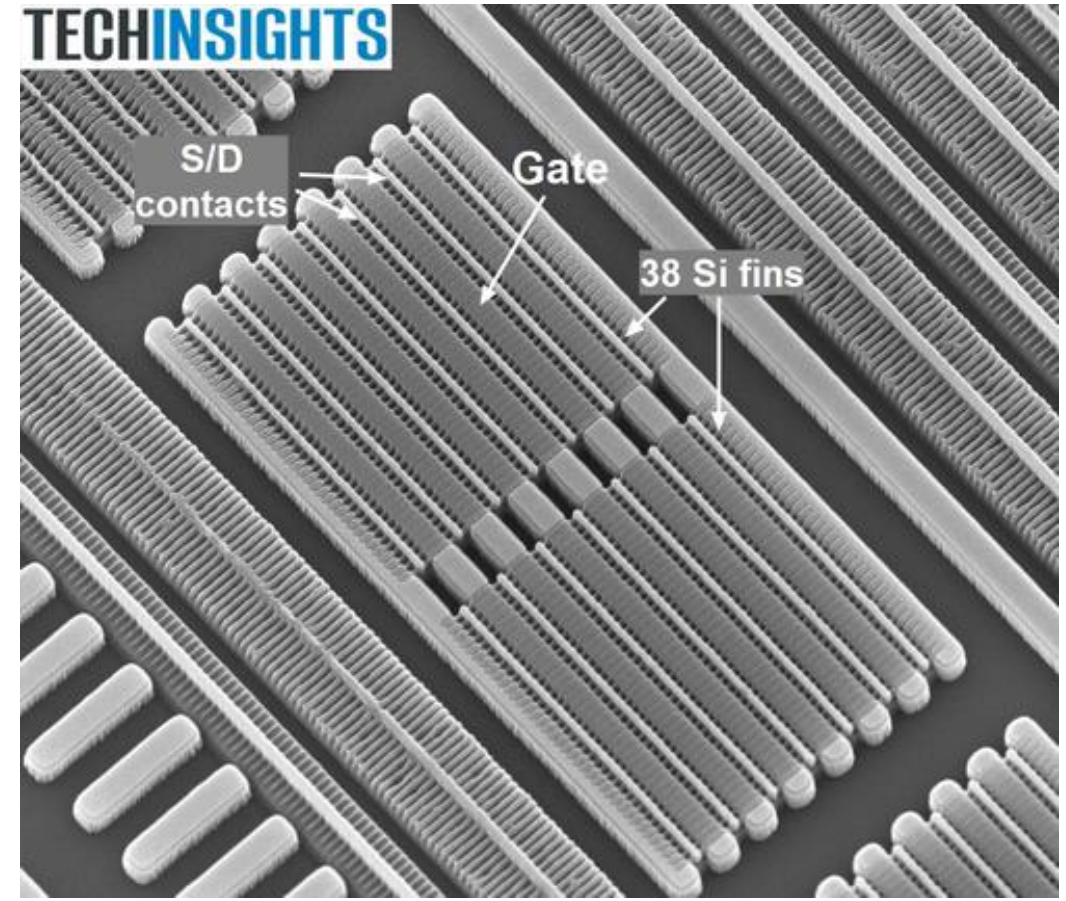
02 Household distribution ~m



03 Printed circuits ~mm



04 Integrated circuits (μm to nm)

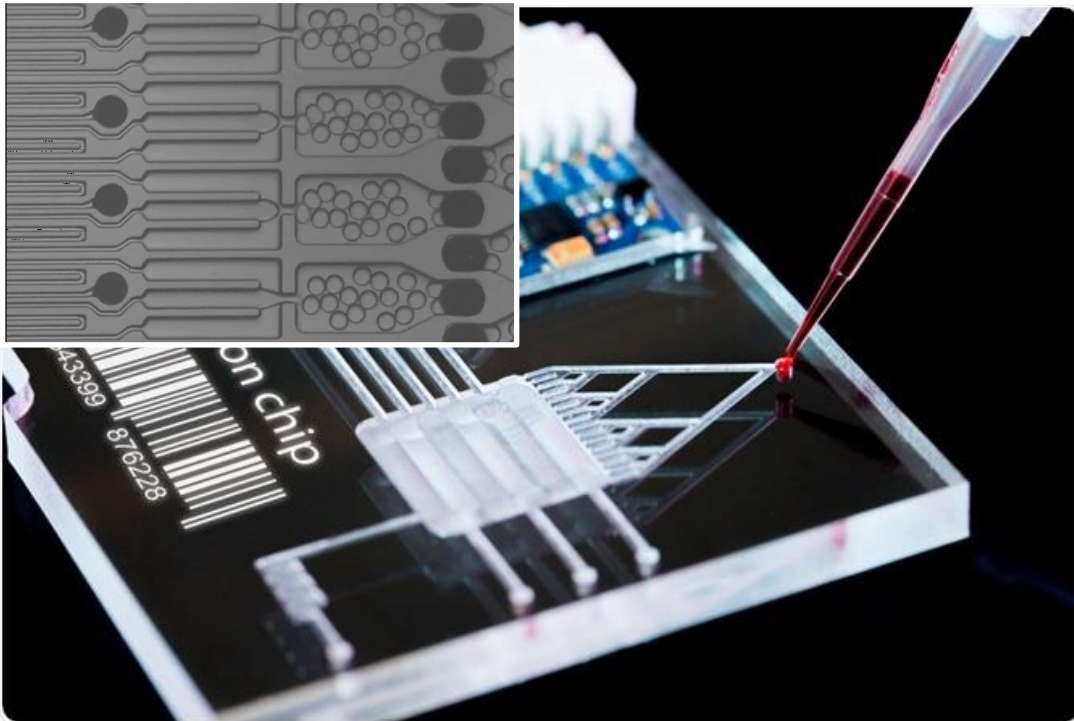


1.2 Electric Circuits (d)

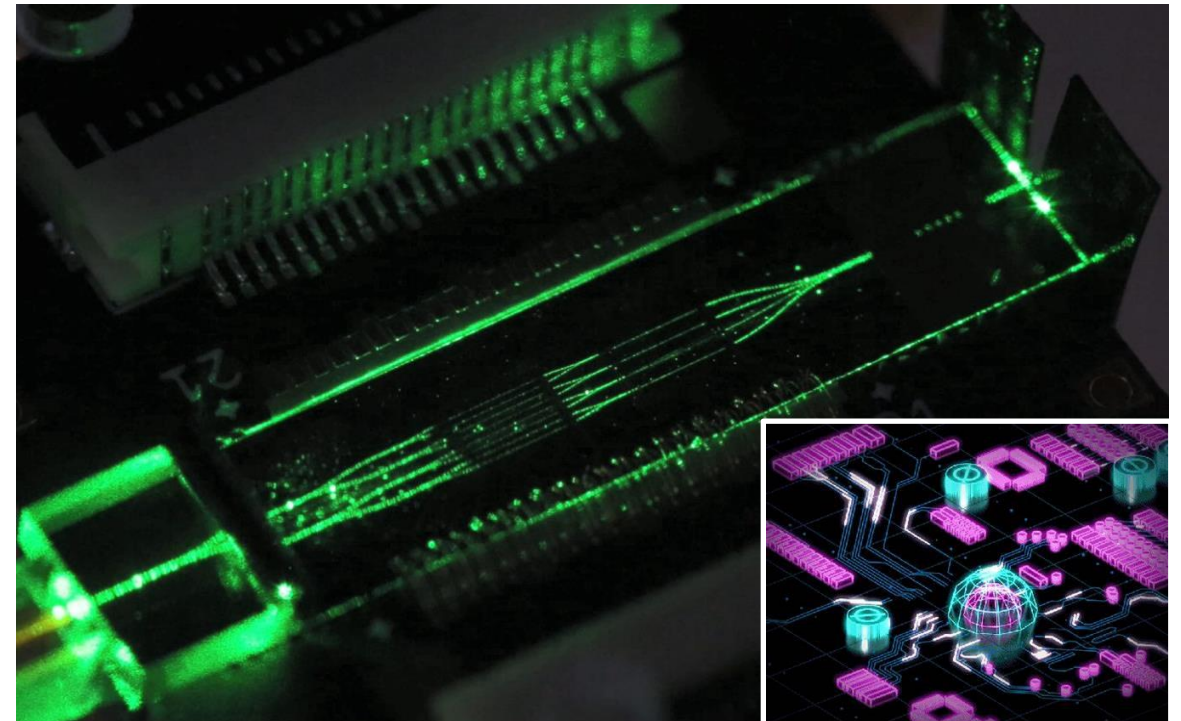
- Other circuits:

a) In this course we focus on electric circuits, yet there are other types of circuits.

Microfluidic circuits



Photonic integrated circuits



Chapter 1 Basic Concepts

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1.3 Charges and Current (a)

- Charges (电荷)

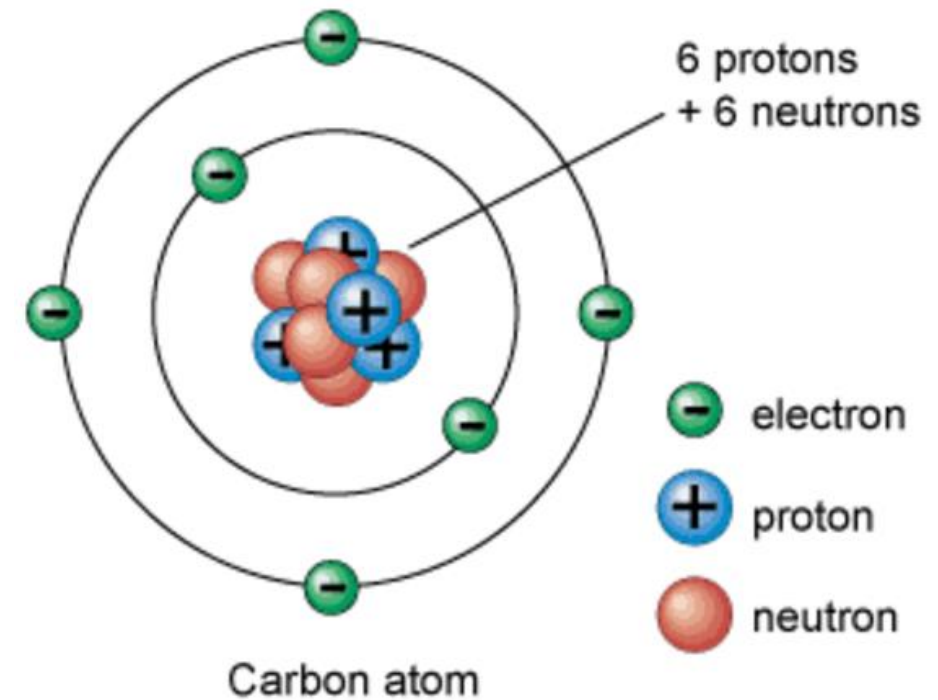
The charge is an electrical property of the atomic particles (electrons, positrons, protons, ions), measured in coulombs (C).

- elementary charge $q = 1.602 \times 10^{-19} \text{ C}$
- Electron: $e = -1.602 \times 10^{-19} \text{ C}$
- Proton or Positron is $+1.602 \times 10^{-19} \text{ C}$

Charge of any matter can only be the multiple of the elementary electron charge q .

- *Coulomb is a large unit, other units like pC, nC, or μC are typically used.*

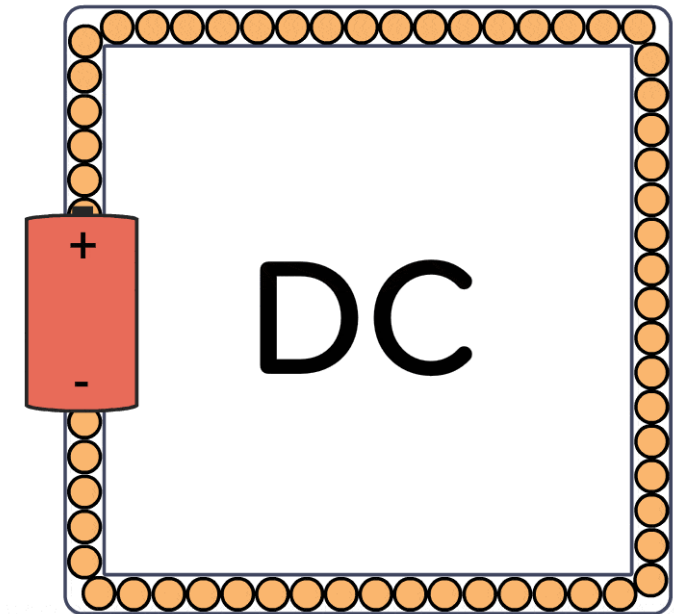
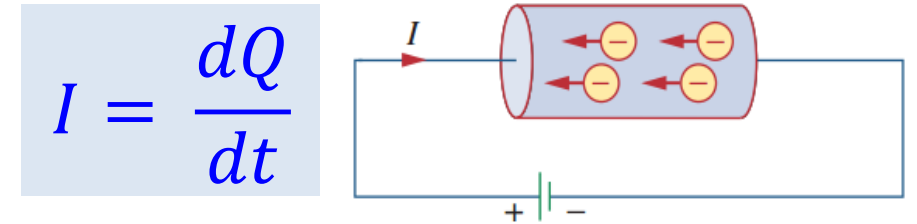
The law of conservation of charge | 电荷守恒定律



1.3 Charges and Current (b)

- Electric current (电流)

- a) A unique feature of electric charge is the fact that it **can be mobile**; that is, it can be transferred from one place to another, where it can be converted to another form of energy.
- b) The **collective movement of charges (电荷的定向移动)** result in **electric current (电流)**.
- c) The direction of electric current is defined for particles of positive charge, that is the opposite of the flow of electrons though physically electrons are moving.
- d) Electric current is defined as **the change of charge per unit time**, measured in **amperes (A)**.

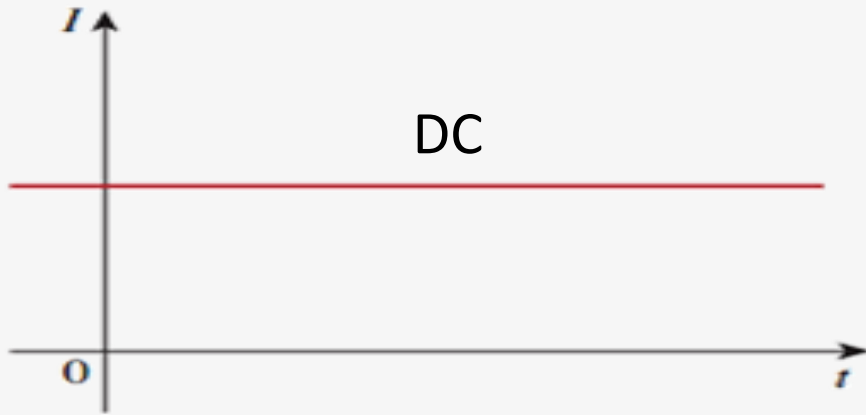


1.3 Charges and Current (c)

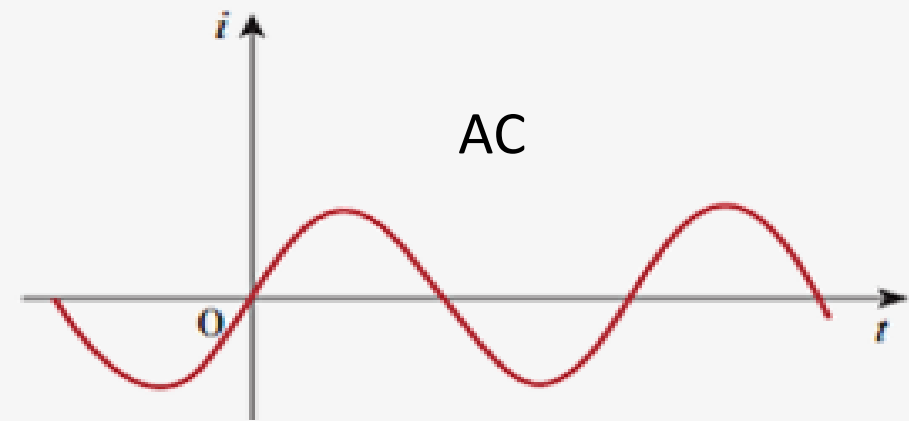
- Types of electric current (电流类型): AC and DC

- The charge transferred between time t_0 and t is given as:
- Current can take various forms against time.

$$Q \triangleq \int_{t_0}^t i \, dt$$



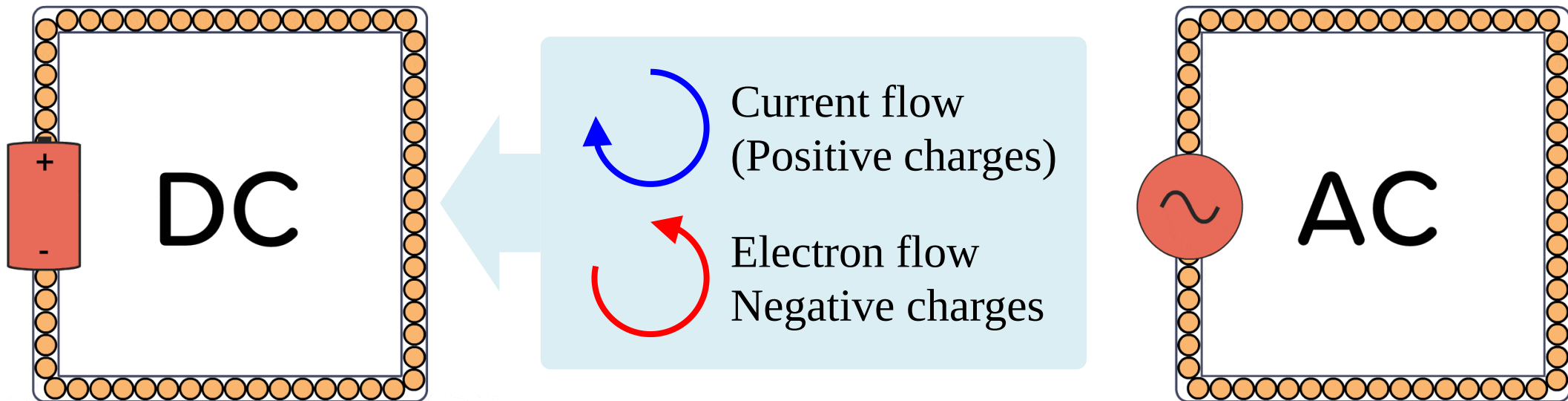
DC | 直流 -- A current that remains constant with time is called Direct Current, conventionally represented by I (in capital). A common source of DC is a battery.



AC | 交流 -- A current that varies sinusoidally with time is called Alternating Current, conventionally represented by i (in lowercase). Wall sockets are an example of AC.

1.3 Charges and Current (d)

- Direction of current flow (电流方向)

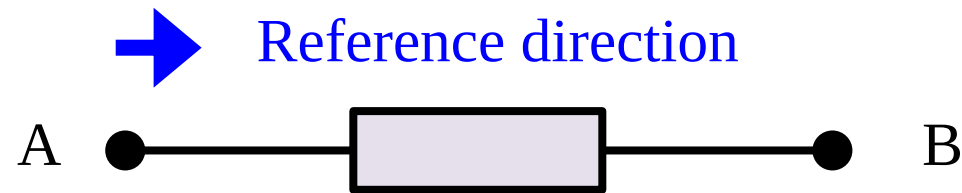


- a) Once we define current as the movement of charge, we expect current to have an associated direction of flow. The direction of current flow is conventionally taken as the direction of positive charge movement, introduced by American scientist Benjamin Franklin (1706–1790), while in many cases only electron flow in the other direction.

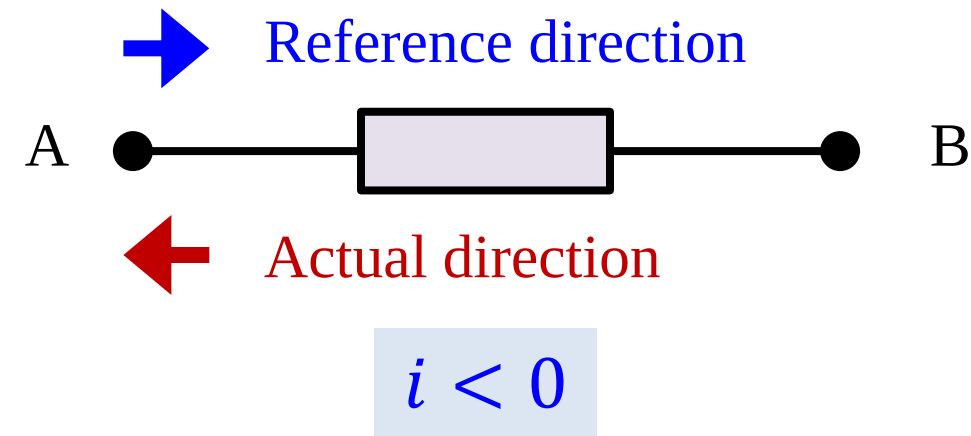
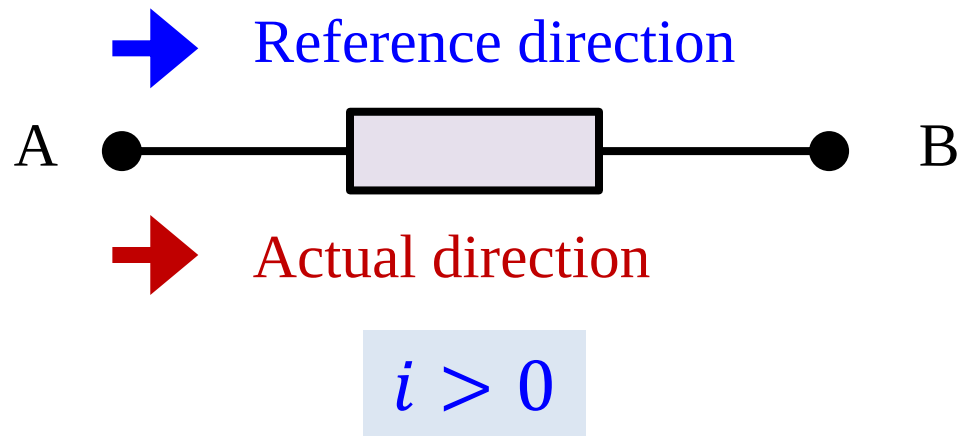
1.3 Charges and Current (e)

- Reference direction of current (参考电流方向)

a) Reference direction: presumed direction that current flows in.



b) Reference and actual directions:





André-Marie Ampère | 安德烈·玛丽·安培

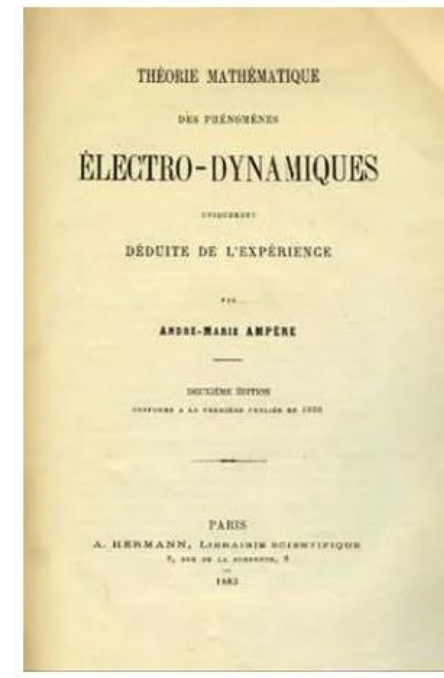
1775.01.20 – 1836.06.10

French Physicist, Chemist and Mathematician

One of the founders of the field of electromagnetism

Interesting
To Know

Year	Accomplishments (selected)
1810	Recognition of the existence of the element Fluorine
1816	Correctly identification that chemical elements should be organized according to their properties
1820	Right-hand grip rule
1823	Ampere's force law
1826	Foundational text of electromagnetism



Ampere's
greatest work –
*Memoir on the
Mathematical
Theory of
Electrodynami
c Phenomena*

1.3 Charges and Current (f)

- Example 01

Example 01: How much charge is represented by 4600 electrons ?

Solution:

Elementary charge $q = 1.6 \times 10^{-19} \text{ C}$.

An electron carries a negative charge, so $e = -q = -1.6 \times 10^{-19} \text{ C}$

4600 electrons will have $Q = -1.6 \times 10^{-19} \text{ C} \times 4600 \text{ electrons} = -7.36 \times 10^{-16} \text{ C}$

Practice Example 01: Calculate the charge represented by 6 million protons.

1.3 Charges and Current (g)

- Example 02

Example 02: The total charge entering a terminal is given by $Q = 5t \sin(4\pi t)$ mC. Calculate current at $t = 0.5$ s

Solution:

$$i = \frac{dq}{dt} = \frac{d}{dt} (5t \sin(4\pi t)) \text{ mC/s} = (5 \sin(4\pi t) + 20 \pi t \cos(4\pi t)) \text{ mA}$$

At $t = 0.5$

$$i = 5 \sin(2\pi) + 10 \pi \cos(2\pi) = 0 + 10 \pi = 31.42 \text{ mA}$$

1.3 Charges and Current (h)

- Example 03

Example 03: Determine the total charge entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is $i = (3t^2 - t)$ A

Solution:

$$i = \frac{dq}{dt},$$

$$Q = \int_{t=1}^2 i dt = \int_1^2 (3t^2 - t) dt = \left(t^3 - \frac{t^2}{2} \right) \Big|_1^2 = (8 - 2) - \left(1 - \frac{1}{2} \right) = 5.5 \text{ C}$$

1.4 Voltage (a)

- Voltage (电压)

- a) To move the electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by an external **electromotive force (EMF)**, typically represented by the battery.
- b) This EMF is also known as **voltage or potential difference**. The voltage V_{ab} between two points a and b in a circuit is the energy (or work) needed to move a unit charge from a to b, divided by q.
- c) Positive charge moving from a higher potential to a lower one **yields energy**, while its movement from a lower to a higher potential **requires energy**.
- d) Voltage (or potential difference) is **measured in volts (V)**, and **often denoted by the letter v or V**.

$$V_{ab} = \frac{dw}{dq}$$



1.4 Voltage (b)

- Voltage (电压)

$$V_{ab} = \frac{dw}{dq}$$

where w is energy in joules (J) and q is charge in coulombs (C). The V_{ab} is measured in volts (V).

1 volt = 1 joule/coulomb = 1 newton-meter/coulomb

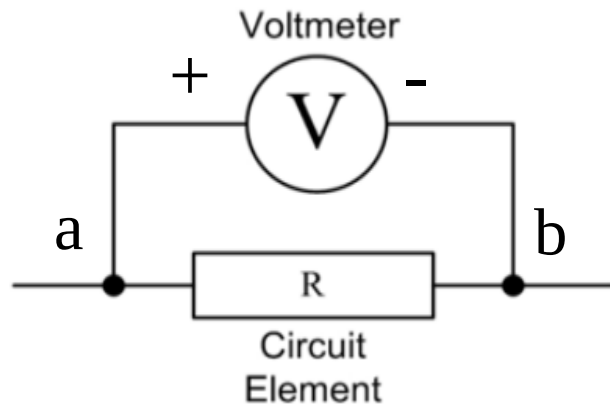


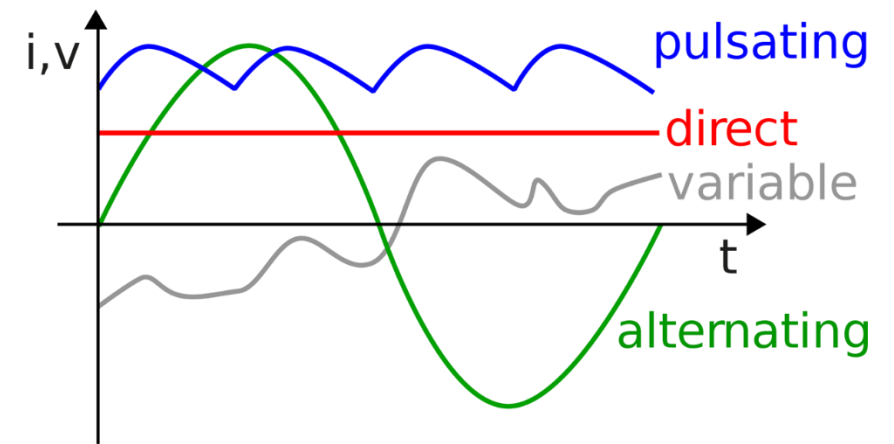
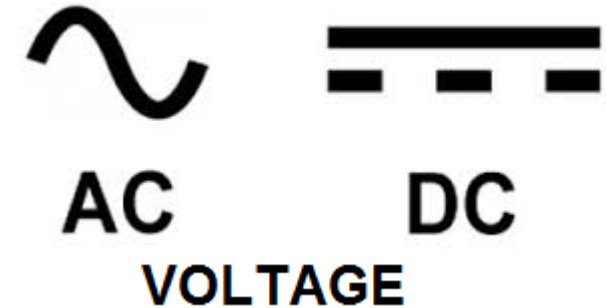
Fig. 1.4

- Voltage is always measured **across a circuit element**
- Figure 1.4 shows the voltage across an element connected to points a and b . The plus (+) and minus (-) signs are used to **define reference direction or voltage polarity**.
- It follows logically that in general $V_{ab} = -V_{ba}$

1.4 Voltage (c)

- Forms of voltage: DC and AC

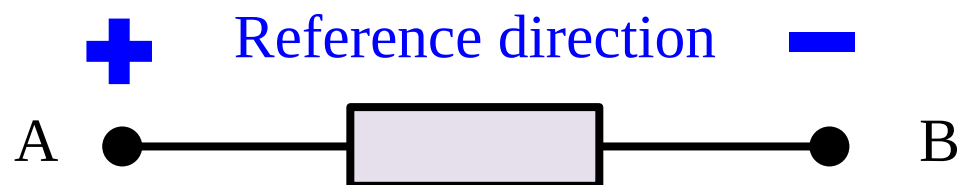
- a) A constant voltage is called a **DC voltage** and is represented by V
- b) A sinusoidally time-varying voltage is called an **AC voltage** and is represented by v .
- c) A DC voltage is commonly produced by a battery; while an AC voltage by an electric generator. Keep in mind that **electric current is always through an element** and that **electric voltage is always across the element or between two points**.



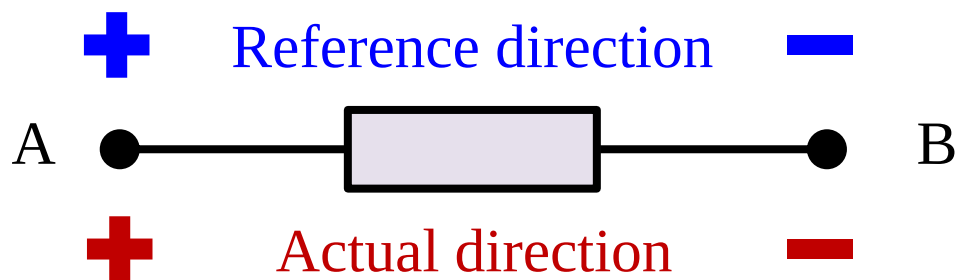
1.4 Voltage (d)

- Reference direction of voltage (参考电压方向)

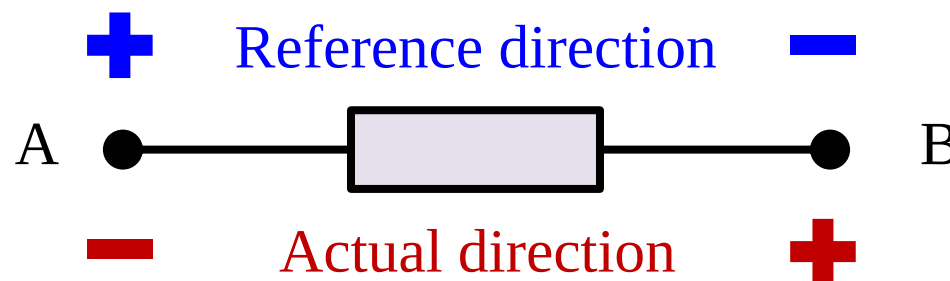
a) Reference direction: presumed direction of voltage.



b) Reference and actual directions:



$$V_{AB} > 0$$



$$V_{AB} < 0$$



Count Alessandro Giuseppe Antonio Anastasio Volta

亚历山德罗·伏特 伯爵

1745.02.18 – 1827.03.05

Italian Physicist and Chemist

Volta's Law of Capacitance & Inventor of the first battery

Interesting
To Know

Year	Accomplishments (selected)
1777	Pioneer in the field of eudiometer (测气管)
1778	The First to discover and isolate methane
1778	Volta's Law of Capacitance (V proportional to Q)
1794	Made the first electrochemical series (电化序)
1800	The first battery (voltaic pile or voltaic column)
1804	Made a count by Napoleon

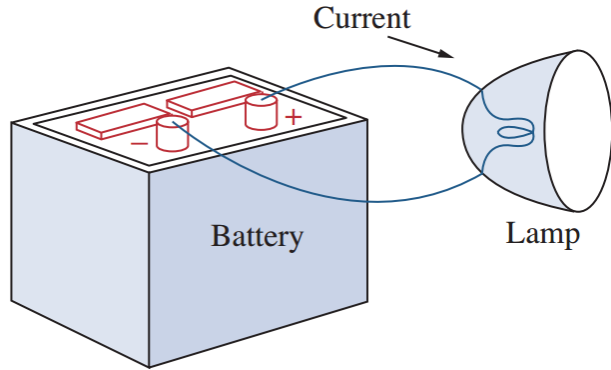


Tempio Voltano in Como, Italy



1.4 Voltage (e)

- Example 04



Example 04: An energy source forces a constant current of 2 A for 10 s to flow through a lightbulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

$$V_{ab} = \frac{dw}{dq}$$

$$I = \frac{dq}{dt}$$

Solution:

$$dq = I \cdot dt = 2 \text{ A} \cdot 10 \text{ s} = 20 \text{ C}$$

$$V = dw/dq = 2300 \text{ J} / 20 \text{ C} = 115 \text{ V}$$

1.5 Power and Energy (a)

- Power and Energy (功率和能量)

$$p = \frac{dw}{dt} \longrightarrow p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi \longrightarrow p = vi$$

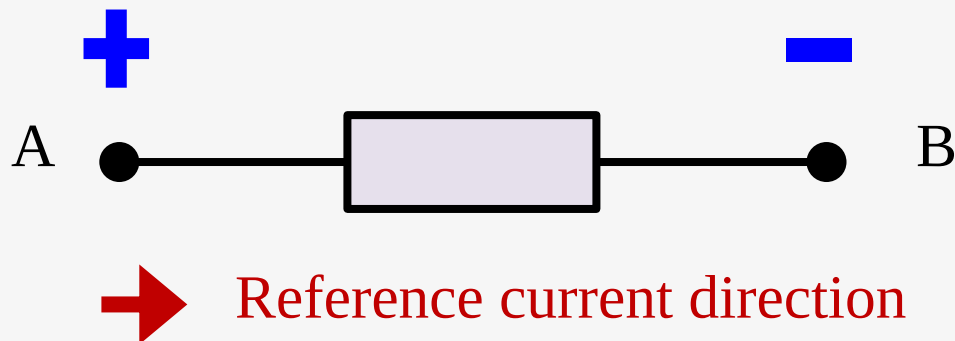
- a) Although current and voltage are the two basic variables in an electric circuit, they are not sufficient by themselves. For practical purposes, we need to know how much power an electric device can handle.
- b) Power (功率) is the time rate of generating or absorbing energy (能量), measured in watts (W).
- c) The power absorbed or supplied by an element is the product of the voltage across the element and the current through it.

1.5 Power and Energy (b)

- Associated (关联) and nonassociated (非关联) reference directions
- a) Associated reference direction (Passive sign convention, 无源符号约定): is satisfied if the direction of current is selected such that current enters the terminal that is more positively biased.
- b) Nonassociated reference direction: is satisfied if the direction of current is selected such that current enters the terminal that is more negative biased.

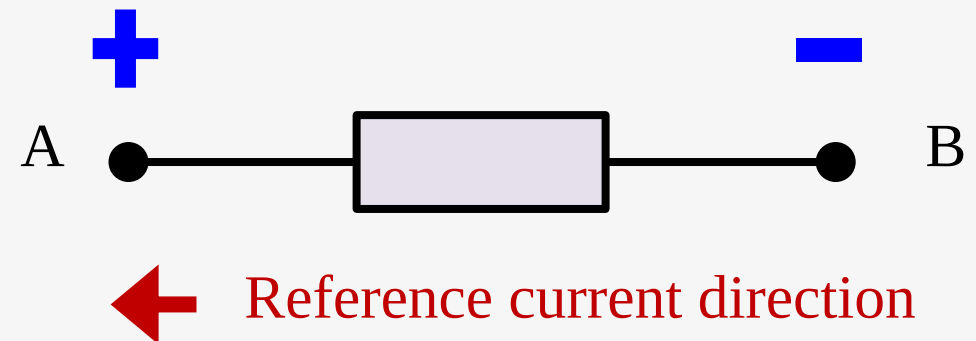
Associated reference directions

Reference voltage direction



Nonassociated reference directions

Reference voltage direction



1.5 Power and Energy (c)

- Associated (关联) and nonassociated (非关联) reference directions

$P = vi > 0$ absorb energy

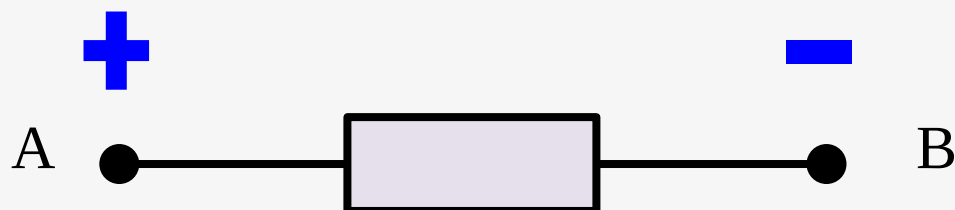
$P = vi < 0$ supply energy

$P = vi > 0$ supply energy

$P = vi < 0$ absorb energy

Associated reference directions

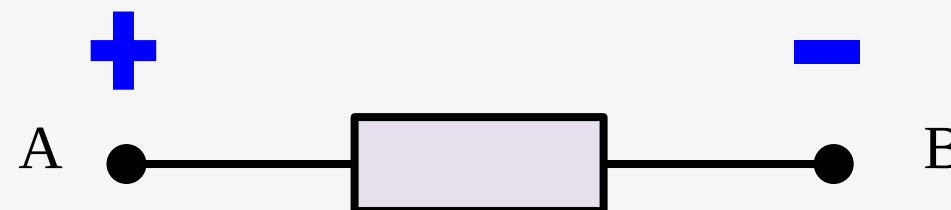
Reference voltage direction



Reference current direction

Nonassociated reference directions

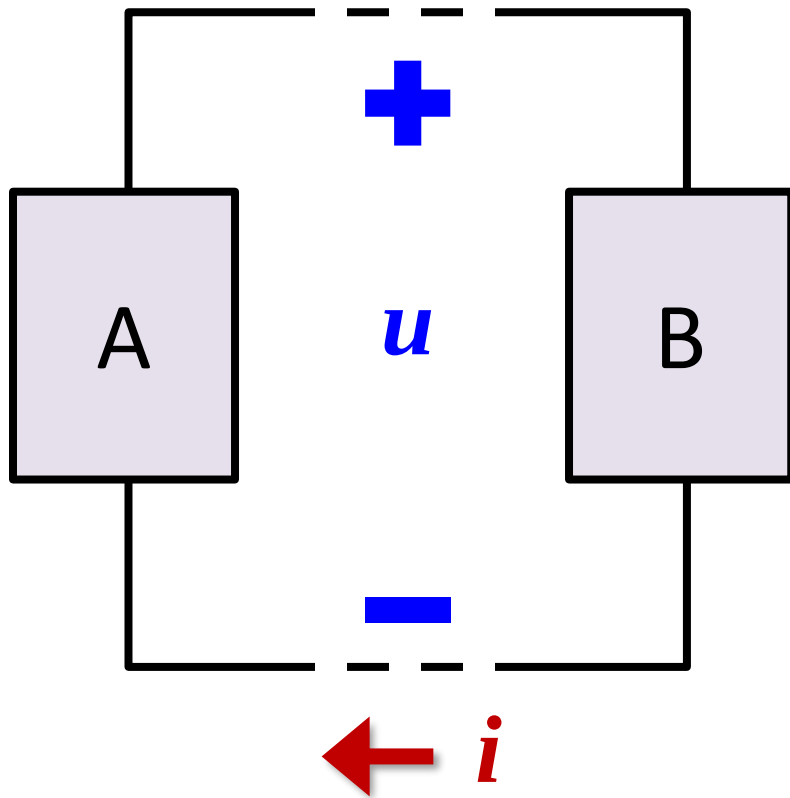
Reference voltage direction



Reference current direction

1.5 Power and Energy (d)

- Example 05



Q1. Device A: associated or nonassociated ?

Q2. Device B: associated or nonassociated ?

Q3. Assuming $u = 5 \text{ V}$ and $i = 2 \text{ A}$. Which device would it supply the power?

1.5 Power and Energy (e)

- The law of conservation of energy (能量守恒定律)
 - a) The law of conservation of energy must be obeyed in any electric circuit, i.e., energy cannot be created or destroyed. Thus power also must be conserved. For this reason, the algebraic sum of power in a circuit, at any instant of time, must be zero.
 - a) The total power supplied to the circuit must balance the total power absorbed.
 - b) The sum of all power supplied must be absorbed by the other elements.
 - b) Energy is the capacity to do work, measured in joules (J). The energy absorbed or supplied by an element from time instant t_0 to t is
$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt$$
 - c) The electric power utility companies measure energy in watt-hours (Wh), where 1 Wh = 3,600 J and 1 kWh = 1 度

1.5 Power and Energy (f)

- Example 06

Example 06: Find the power delivered to an element at $t = 3$ ms if the current entering its positive terminal is $i = 5\cos(60\pi t)$ A, and the voltage is: (a) $v = 3i$, (b) $v = 3di/dt$

Solution:

(a) The voltage is $v = 3i = 15\cos(60\pi t)$ V, hence, the power is

$$p = vi = 75\cos^2(60\pi t) \text{ W}$$

At $t = 3 \text{ ms} = 3 \times 10^{-3} \text{ s}$,

$$p = 75\cos^2(60\pi \times 3 \times 10^{-3}) = 75\cos^2(0.18\pi) = 53.48 \text{ W}$$

(b) We find the voltage and the power as

$$v = 3 di/dt = 3(-60\pi)5\sin(60\pi t) = -900\pi\sin(60\pi t) \text{ V}$$

$$p = vi = -4500\pi\sin(60\pi t)\cos(60\pi t) \text{ W}$$

At $t = 3 \text{ ms} = 3 \times 10^{-3} \text{ s}$,

$$p = -4500\pi\sin(0.18\pi)\cos(0.18\pi) = -6396 \text{ W}$$

1.5 Power and Energy (g)

- Example 07

Example 07: How much energy does a 100 W electric bulb consume in two hours

Solution:

$$t = 2 \text{ h} = 120 \text{ min} = 7200 \text{ s}$$

$$w = pt = 100 \times 7200 = 720000 \text{ J} = 720 \text{ kJ}$$

$$\text{This is the same as } w = pt = 100 \text{ W} \times 2 \text{ h} = 200 \text{ Wh}$$

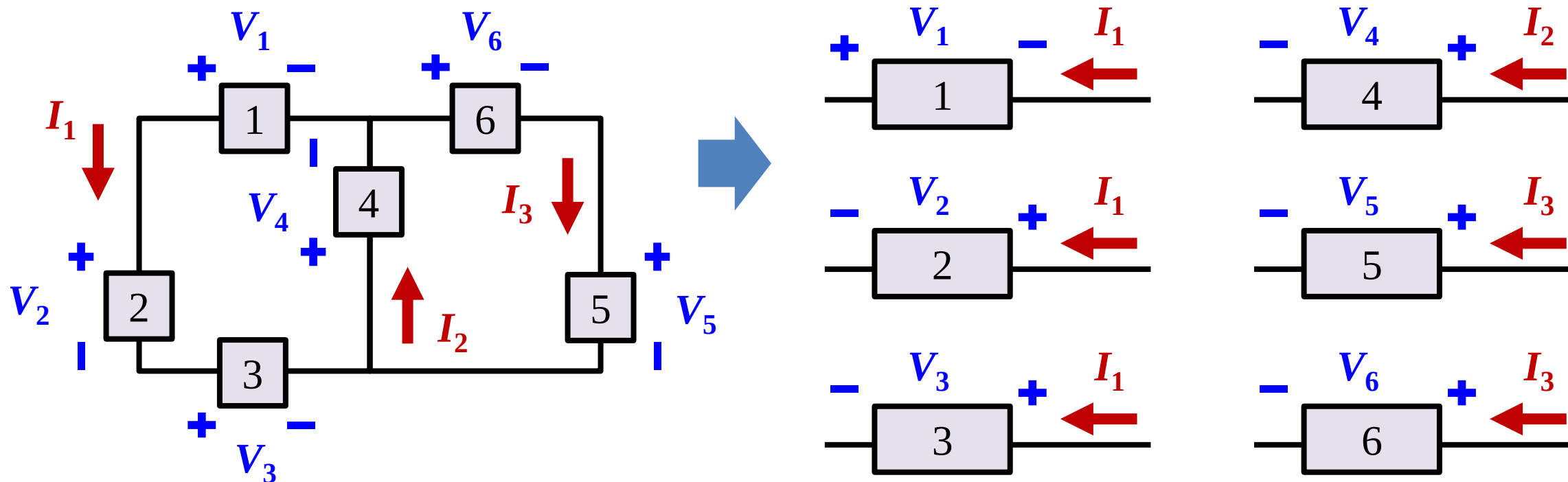
Practice Example 06: A stove element draws 15 A when connected to a 240 V DC line. How long does it take to consume 180 KJ?

1.5 Power and Energy (h)

- Example 08

Example 08: Calculate the power of each element, absorbs or supplies?

$V_1 = 1 \text{ V}$, $V_2 = -3 \text{ V}$, $V_3 = 8 \text{ V}$, $V_4 = -4 \text{ V}$, $V_5 = 7 \text{ V}$, $V_6 = -3 \text{ V}$, $I_1 = 2 \text{ A}$, $I_2 = 1 \text{ A}$, $I_3 = -1 \text{ A}$

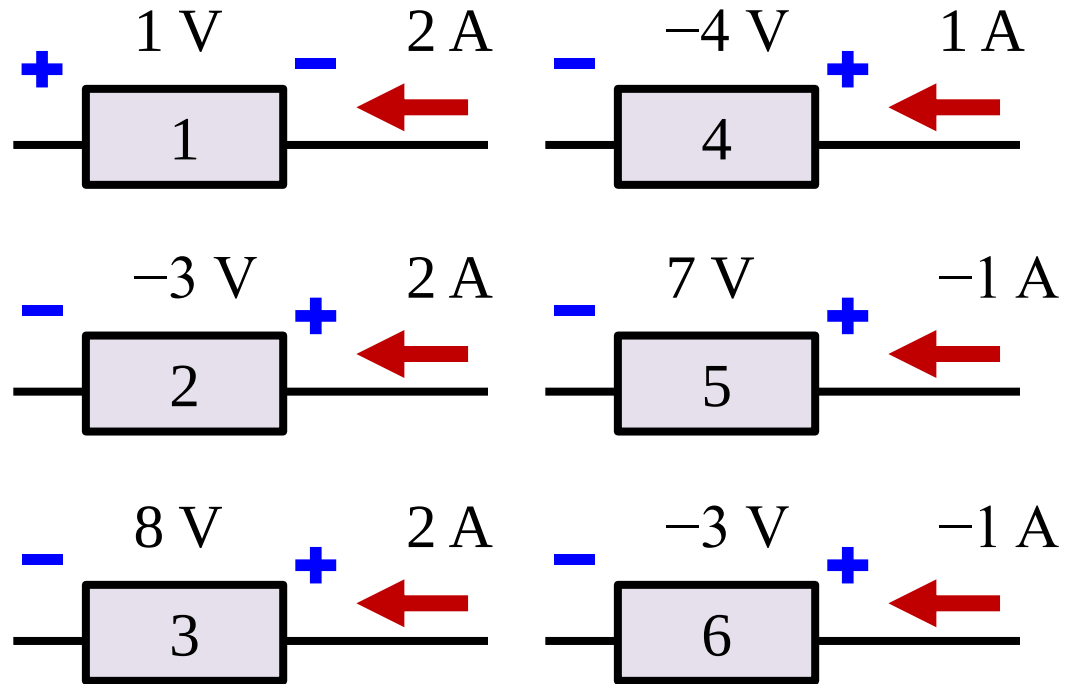


1.5 Power and Energy (h)

- Example 08

Example 08: Calculate the power of each element absorbs or supplies.

$V_1 = 1\text{ V}$, $V_2 = -3\text{ V}$, $V_3 = 8\text{ V}$, $V_4 = -4\text{ V}$, $V_5 = 7\text{ V}$, $V_6 = -3\text{ V}$, $I_1 = 2\text{ A}$, $I_2 = 1\text{ A}$, $I_3 = -1\text{ A}$



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1.6 System of units (a)

- International systems of units (SI)

- As electrical engineers, we deal with **measurable quantities**; when taking measurements, we need to use units to quantify values.
- We adopt the International Systems of Units (SI for short)

Six basic SI units and one derived unit relevant to this text.

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Charge	coulomb	C

1.6 System of units (b)

- The SI prefixes

a) An SI prefix (also known as a metric prefix) is a name or associated symbol that precedes a unit of measure (or its symbol) to form a decimal multiple or submultiple.

b) With prefixes, SI units allow for conveniently representing a wide range of values.

c) $600,000,000 \text{ mm} = 600,000 \text{ m} = 600 \text{ km}$

Prefix	Symbol for Prefix	Scientific Notation
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
hecto	h	10^2
deka	da	10^1
----	--	10^0
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
fermto	f	10^{-15}
atto	a	10^{-18}

1.6 System of units (b)

- Calculation of prefixes

a) In high-power power electronics:

$$I: \text{kA} (10^3 \text{ A}) \quad V: \text{kV} (10^3 \text{ V}),$$

$$P = \text{kA} \times \text{kV} = \text{kk} \times \text{VA} = 10^6 \text{ W} = \text{MW}$$

b) In some integrated circuits:

$$I: \text{nA} (10^{-9} \text{ A}) \quad V: \text{V} (10^0 \text{ V}),$$

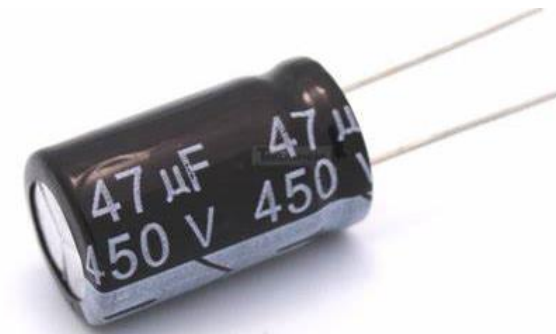
$$P = \text{nA} \times \text{V} = \text{n} \times \text{VA} = 10^{-9} \text{ W} = \text{nW}$$

c) 1 F (capacitance) is a large quantity, therefore we can use nF (10^{-9} F), μF (10^{-6} F)

Power diodes: 8 kV + 2.3 kA



Capacitor: 47 μF



1.6 System of units (c)

- Examples 08

1 mV is equal to 10^{-6} V.

☐ Right

☒ Wrong

The prefix “ μ ” means 10^6 .

☐ Right

☒ Wrong

2,000,000 V is 2 GV.

☐ Right

☒ Wrong

1 GW is equal to 10^9 W

☒ Right

☐ Wrong

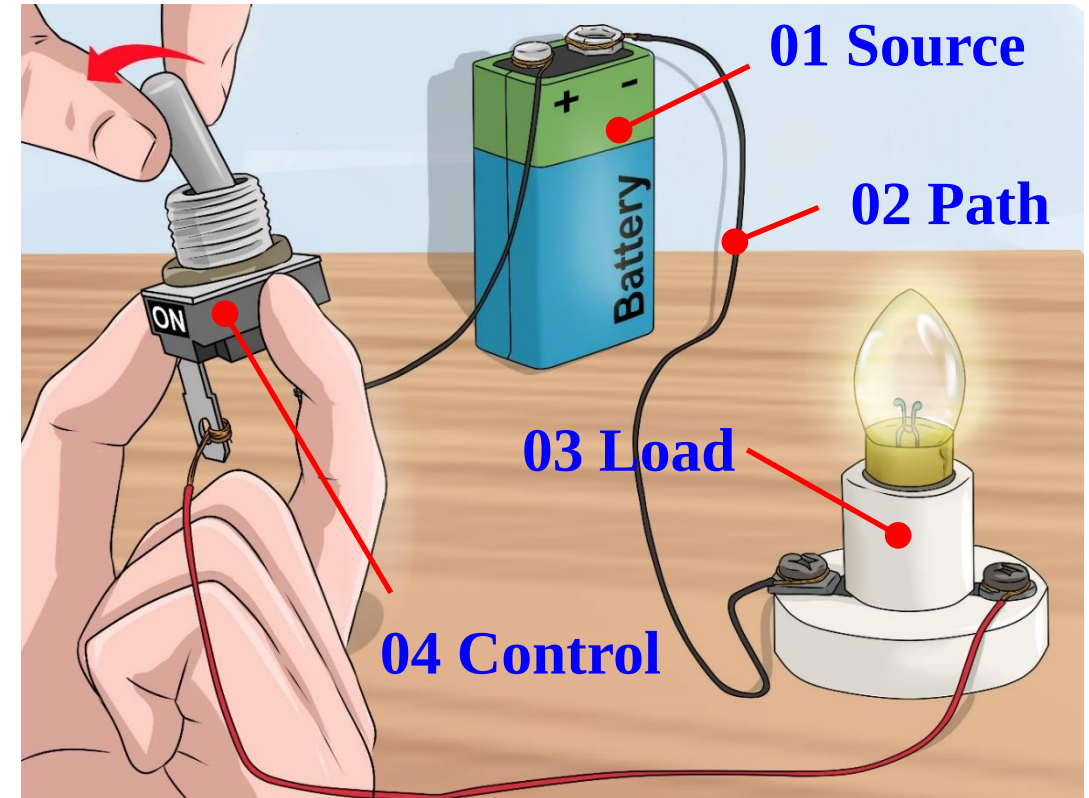
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1.7 Circuit elements (a)

- Elements | components (元件)

- a) An element is the basic building block of a circuit. An electric circuit is simply an interconnection of the elements.
- b) Source + path + load + control
- c) Circuit analysis is the process of determining voltages across (or the currents through) the elements of the circuit.



1.7 Circuit elements (b)

- Typical circuit elements (常见元件)



Resistors



Variable resistors



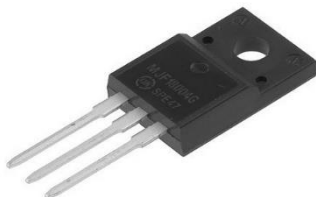
Capacitors



Inductors



Light-dependent resistors



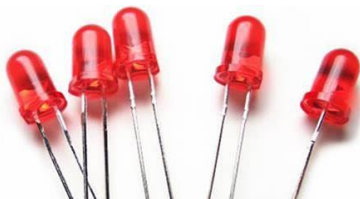
Transistors



Diodes



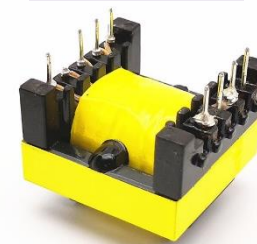
Integrated circuits



LEDs



Thermistors



Transformers



Switches

1.7 Circuit elements (c)

- Active (有源) and passive (无源) elements

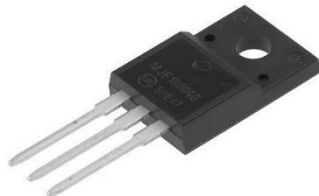
Active Elements control the charge flow in circuits.



Batteries



Generators



Transistors



Integrated circuits

Passive elements do not require external voltage/current to work.



Resistors



Variable resistors



Capacitors



Inductors



Photoresistors



Thermistors



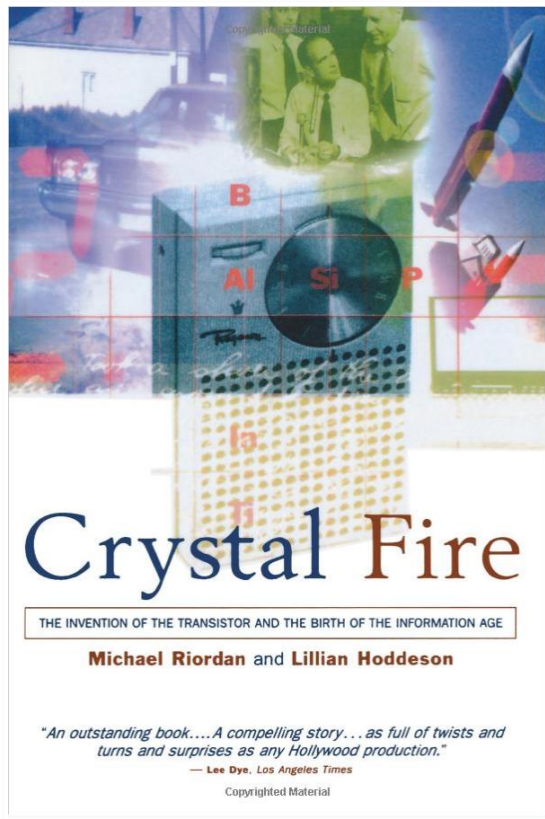
Transformers



Switches

1.7 Circuit elements (d)

- Semiconductor devices



Crystal Fire: The Invention of The Transistor and The Birth of The Information Age

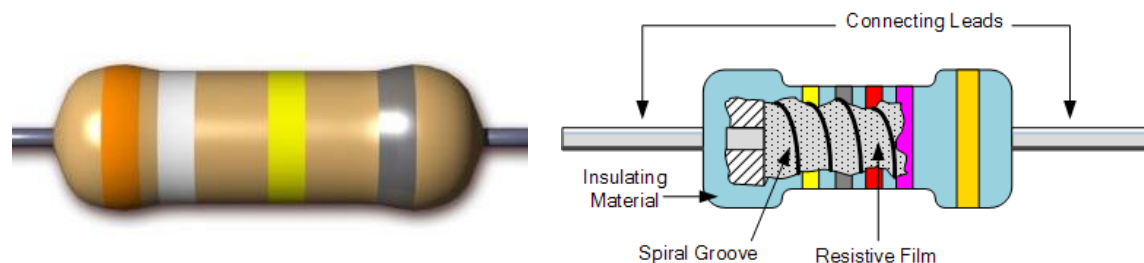
- Michael Riordan & Lillian Hoddeson

“An outstanding book... A compelling story... as full of twists and turns and surprises as any Hollywood production.”

- Lee Dye, Los Angeles Times

1.8 Resistors (a)

- Resistors and resistance (电阻)



Resistors



Resistor symbols

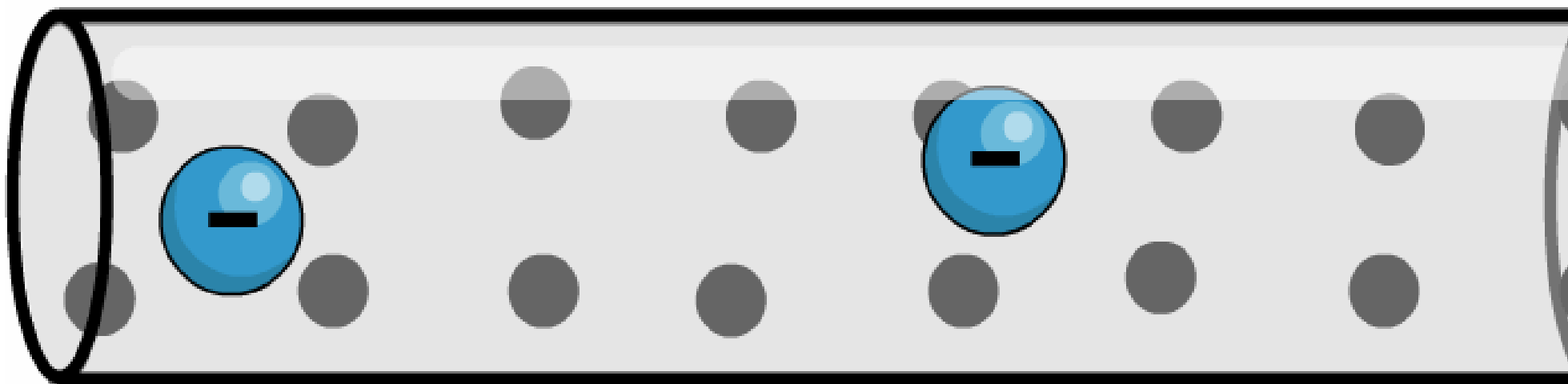
新华字典：(1) 导体对电流通过的阻碍作用 (resistance)
(2) 利用这种阻碍作用做成的元件 (resistors)

Merriam-Webster dictionary:

- (1) **Resistance**: the opposition offered by a body or substance to the passage through it of a steady electric current
- (2) **Resistors**: a device that has electrical resistance

1.8 Resistors (b)

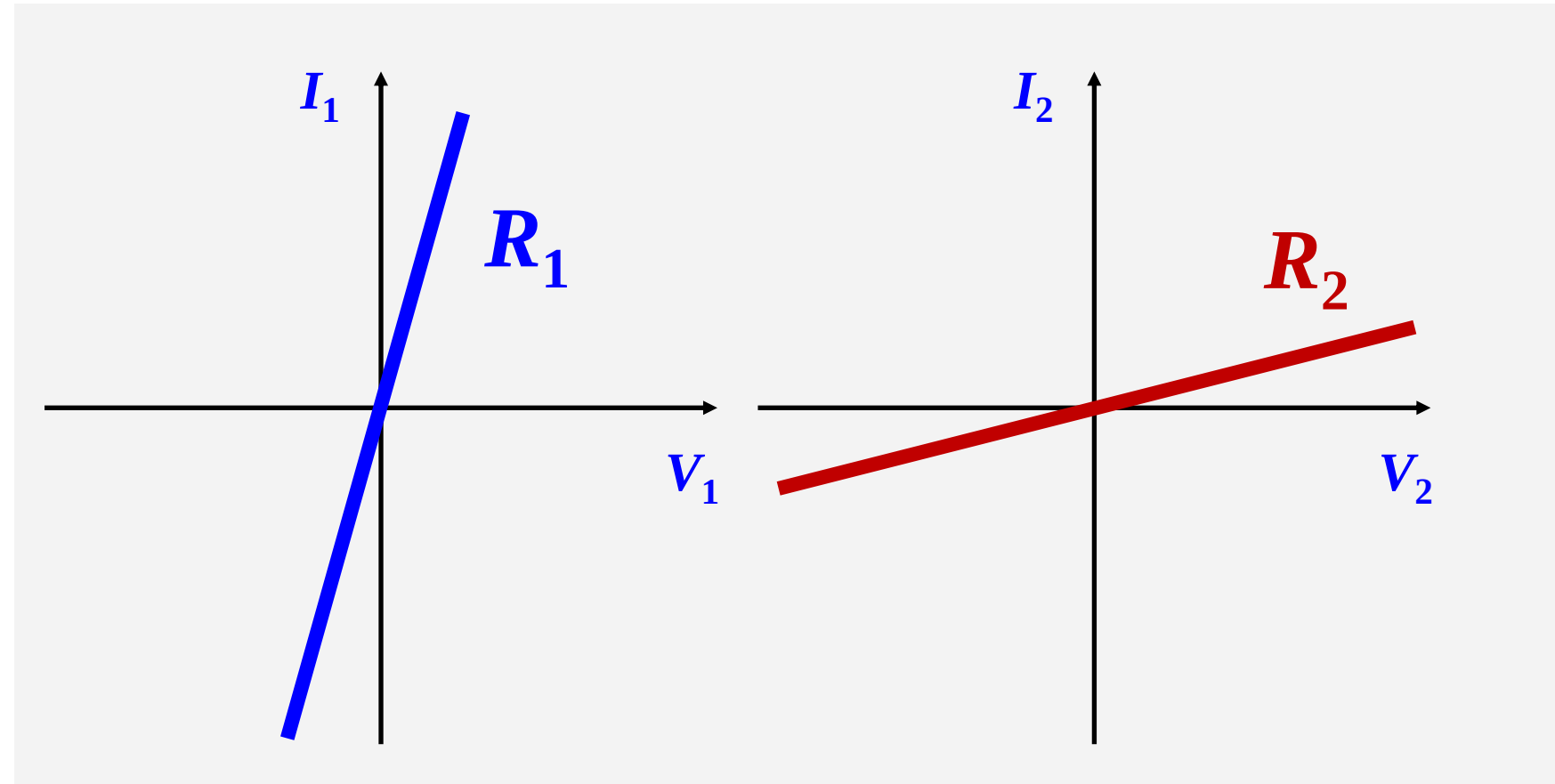
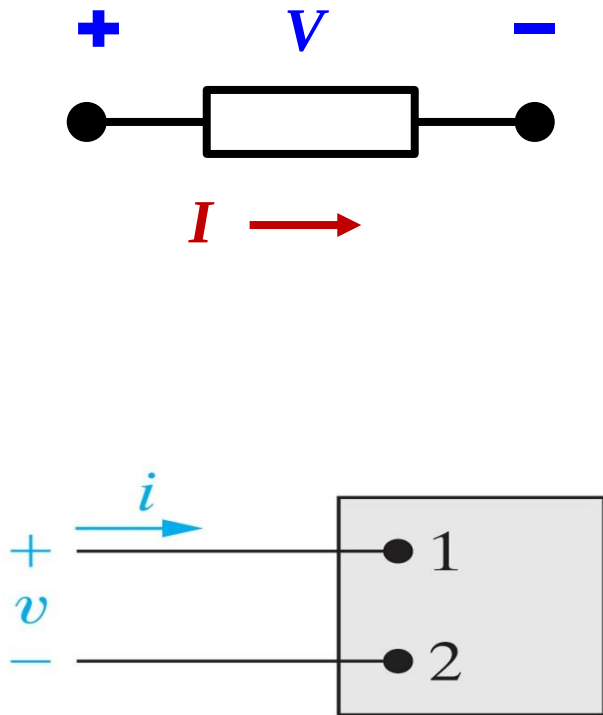
- Resistance (电阻)



Time: 0 seconds

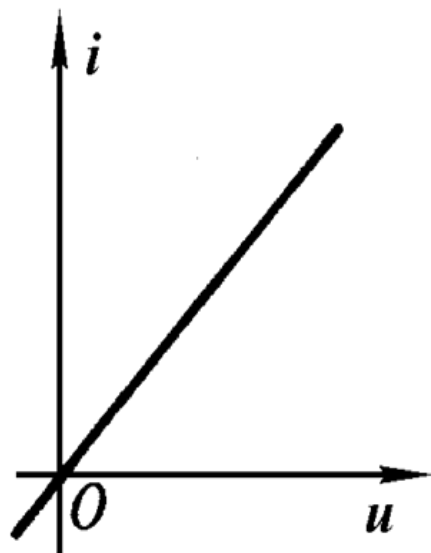
1.8 Resistors (c)

- I - V characteristics (伏安特性曲线)



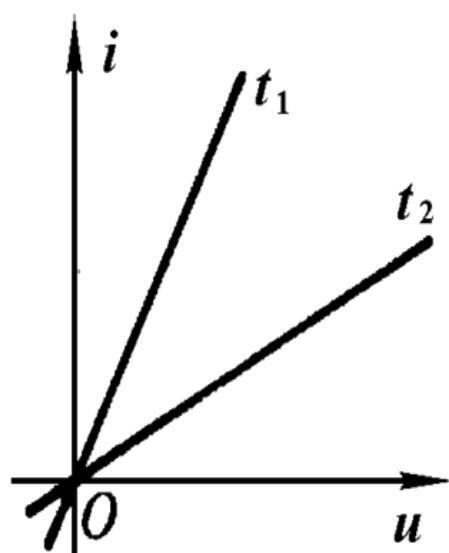
1.8 Resistors (d)

- Types of resistors



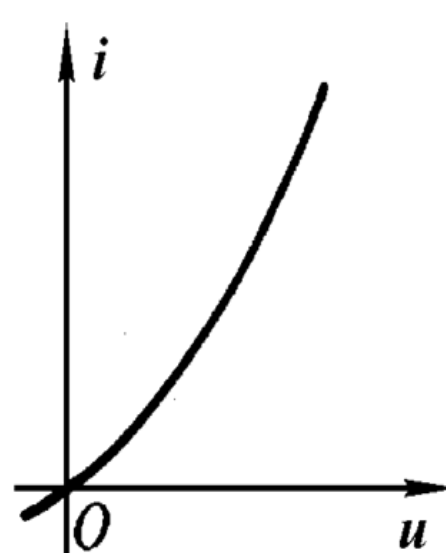
(a)

Linear
time-invariant
resistance



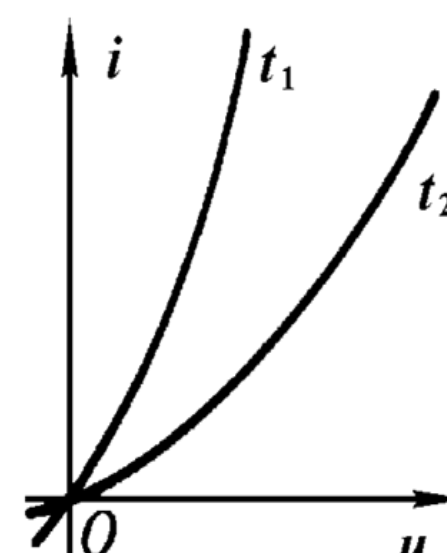
(b)

Linear
time-variant
resistance



(c)

Non-linear
time-invariant
resistance



(d)

Non-linear
time-variant
resistance

1.9 Sources (a)

- Types of sources: dependent and independent

The most important active elements are voltage and current sources (电压源, 电流源) that deliver power to the circuit connected to them. There are two types for each of them: **independent** (独立) and **dependent** (受控).

An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of any other elements in the circuit.



Batteries



Generators



Solar panels

An **ideal dependent (or controlled) source** is an active element in which the source quantity is controlled by another voltage or current in the circuit.



Transistors

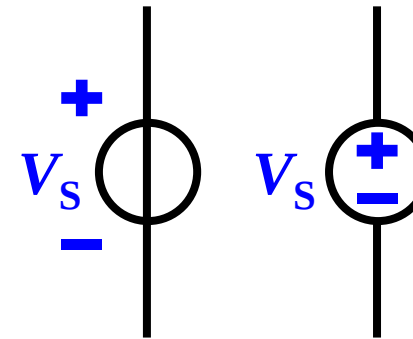


Integrated circuits

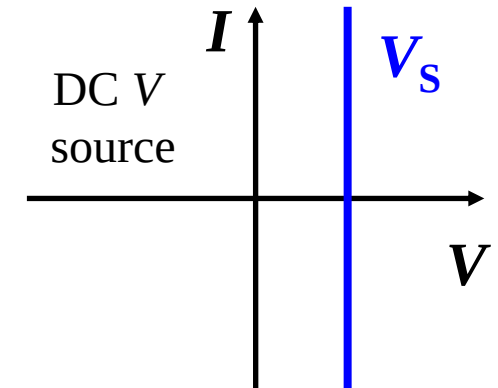
1.9 Sources (b)

- Independent sources (独立电源)

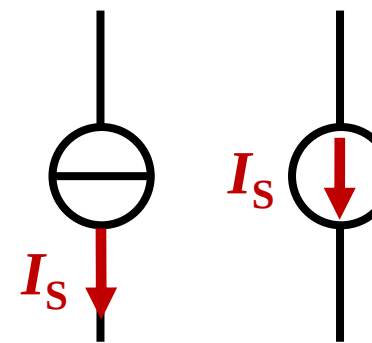
a) An ideal voltage source has zero internal resistance and is capable of producing any amount of current. While the V at its terminals is specified, the I through it is determined by other connected elements.



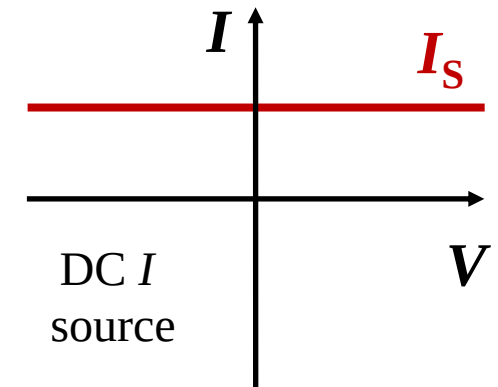
(a) What if $V_s = 0$



b) An ideal current source has infinite resistance. It is able to generate any V to establish the desired I through it. We can know the I through it in advance, but not the voltage.



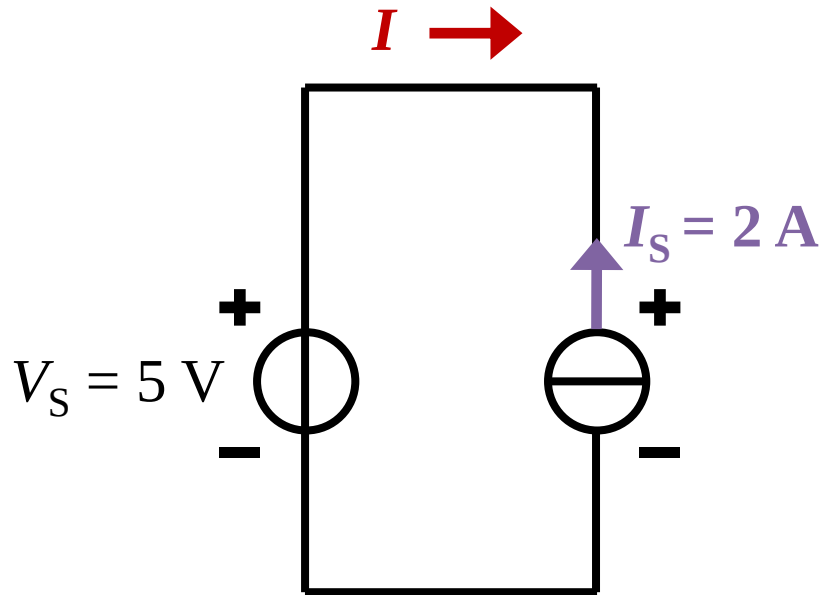
(b) What if $I_s = 0$



1.9 Sources (c)

- Example 09

Example 09: Calculate the power that each element absorbs or supplies.



Solution:

$$I = -I_S = -2\text{ A}$$

$$P_{V\text{-source}} = -2\text{ A} \times 5\text{ V} = -10\text{ W}, \text{ absorbing electricity}$$

(Nonassociated Ref. directions)

$$P_{I\text{-source}} = -2\text{ A} \times 5\text{ V} = -10\text{ W}, \text{ supplying electricity}$$

(Associated Ref. directions)

$$P_{\text{absorb}} = P_{\text{supply}}$$

1.9 Sources (d)

- Dependent sources

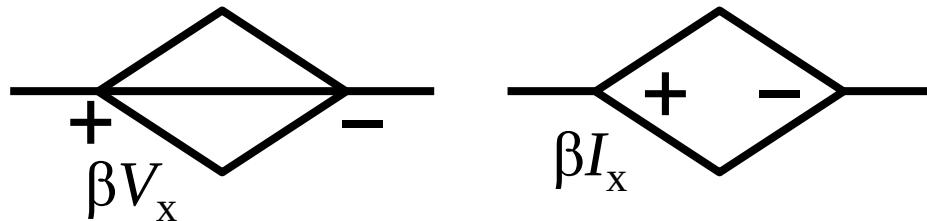
a) An **ideal** dependent (or controlled) source is an active element in which the source quantity is **controlled by another voltage or current in the circuit**, designated by diamond-shaped symbols. There are 2 types of dependent sources:

Dependent voltage sources (受控电压源)

Voltage-controlled voltage sources (VCVS)

Current-controlled voltage sources (CCVS)

$$V_S = \beta V_x \text{ or } V_S = \beta I_x$$

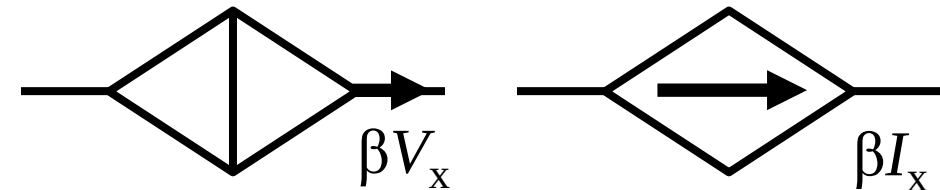


Dependent current sources (受控电流源)

Voltage-controlled current sources (VCCS)

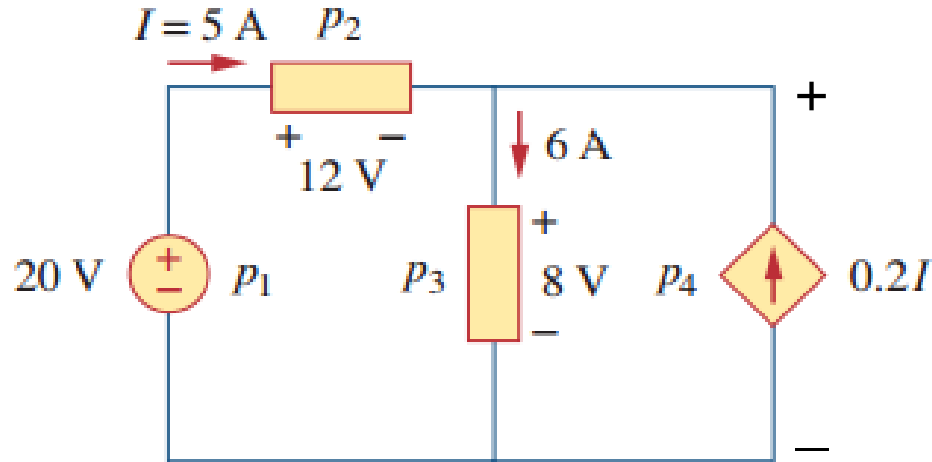
Current-controlled current sources (CCCS)

$$I_S = \beta V_x \text{ or } I_S = \beta I_x$$



1.9 Sources (e)

- Example 10



Example 11: Calculate the power supplied or absorbed by each element.

What else can you learn from this example?

Solution I: use actual directions as references

	V (V)	I (A)	P (W)	Associated or Nonassociated	Absorb or supply?
1	20	5	100	Nonassociated	Supply
2	12	5	60	Associated	Absorb
3	8	6	48	Associated	Absorb
4	8	$0.2 \times 5 = 1$	8	Nonassociated	Supply

Solution II: force all to associated sign convention

	V (V)	I (A)	P (W)	Associated or Nonassociated	Absorb or supply?
1	-20	5	-100	Associated	Supply
2	12	5	60	Associated	Absorb
3	8	6	48	Associated	Absorb
4	8	-1	-8	Associated	Supply

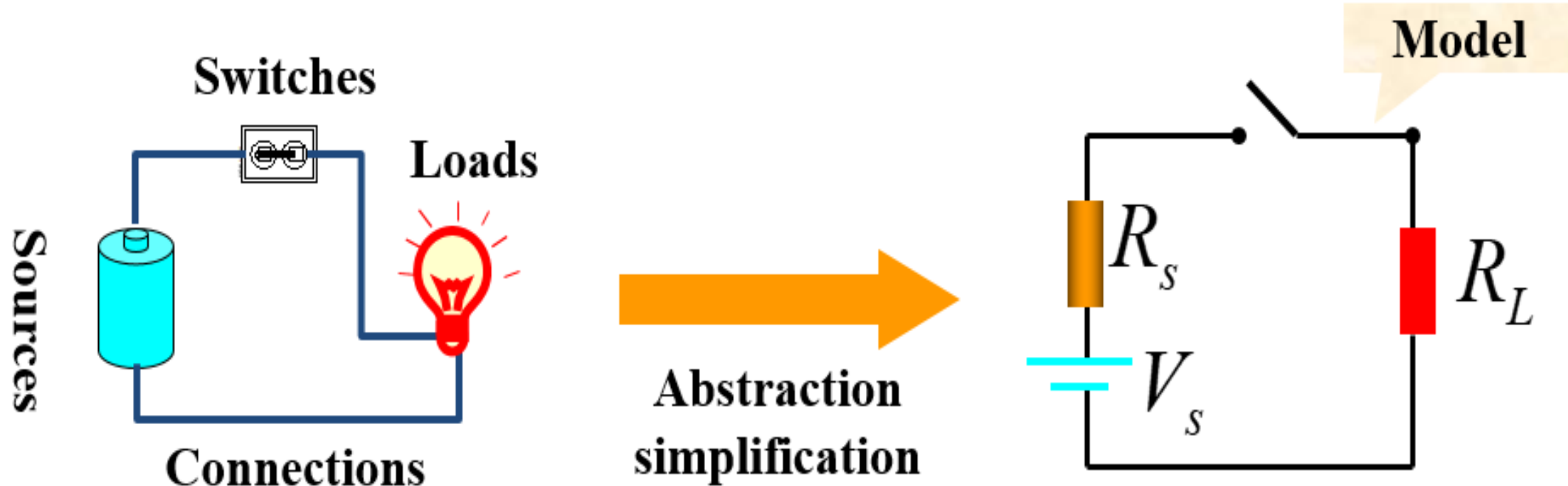
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1.10 Circuit schematics (a)

- Abstraction towards circuit models: a must to enjoy a happy life

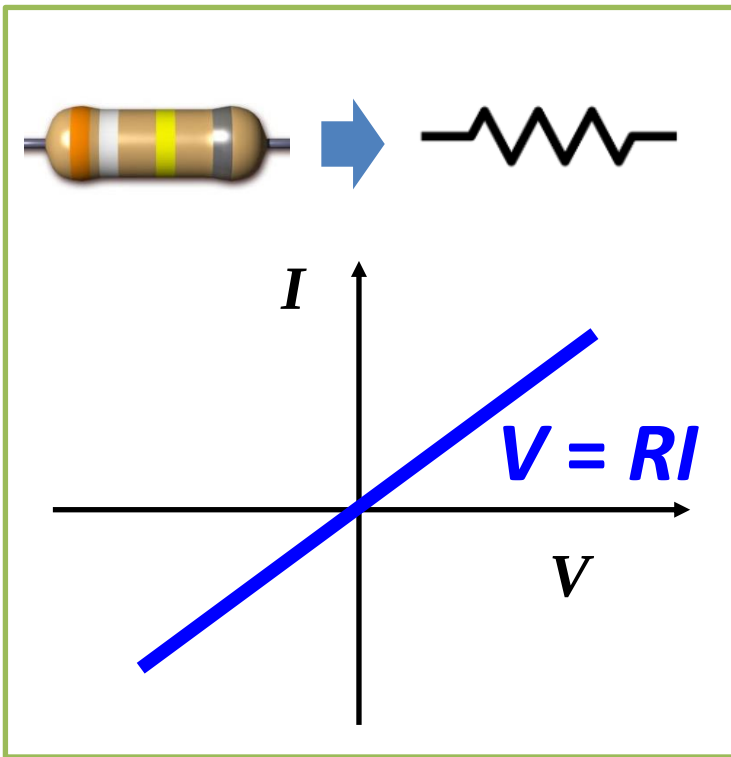
Elements become symbols with idealized mathematical model, which are connected by idea wires and controlled by ideal switches.



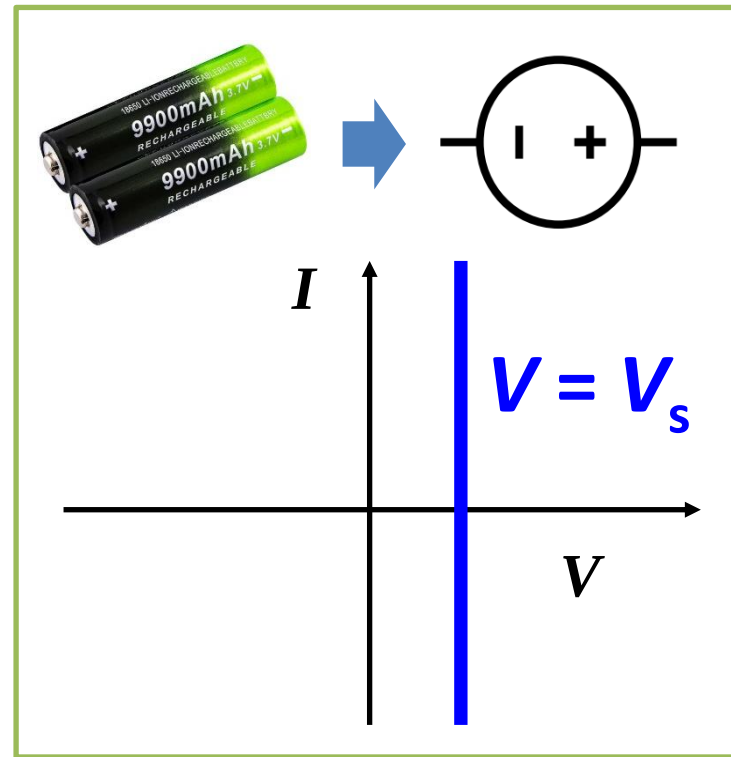
1.10 Circuit schematics (c)

- We have talked about some symbols and their idealized mathematical models:

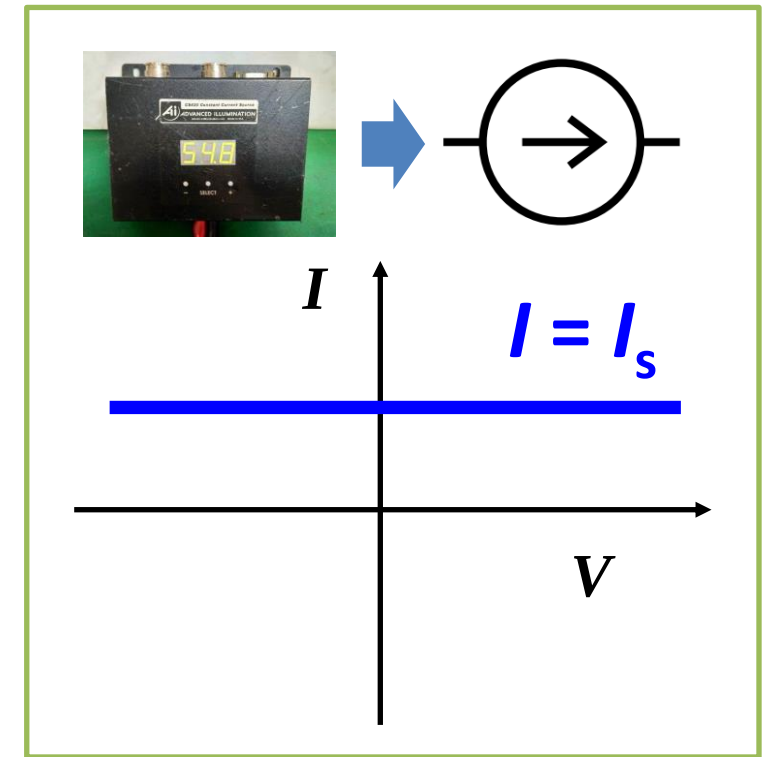
Resistors



Constant V source

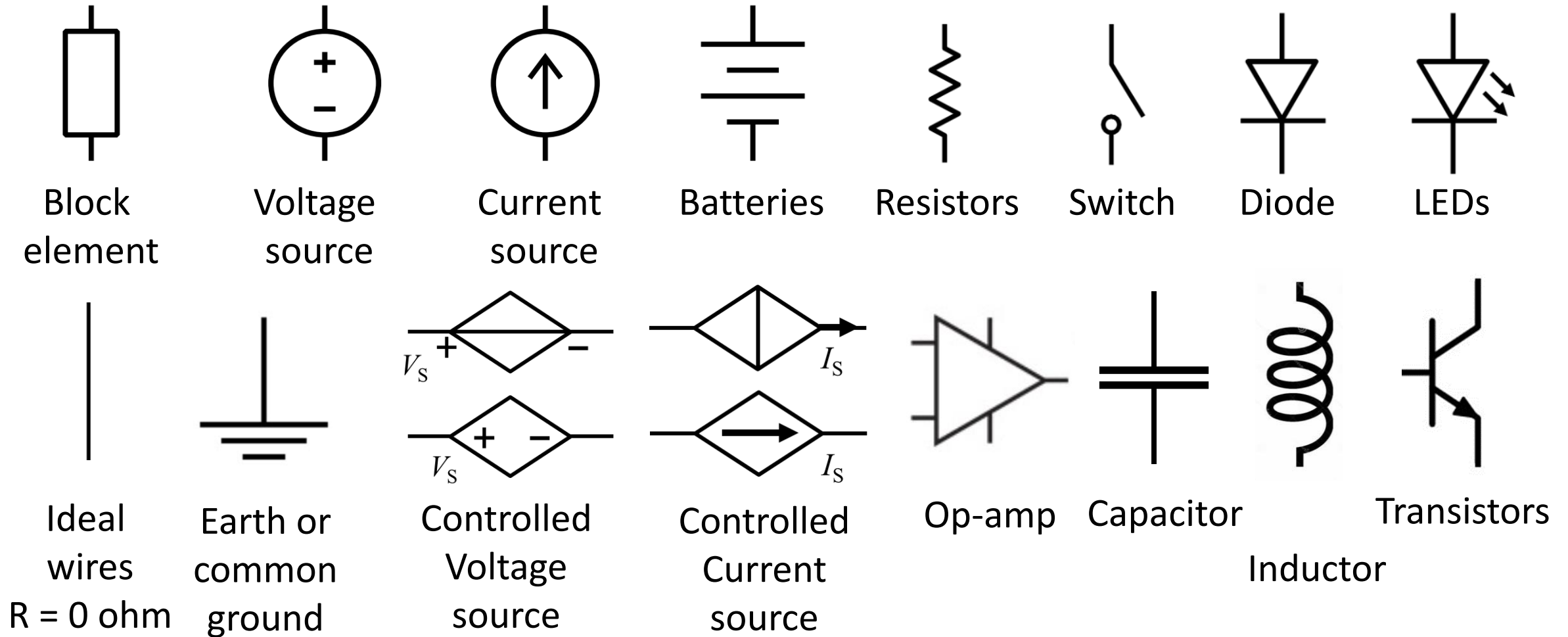


Constant I source



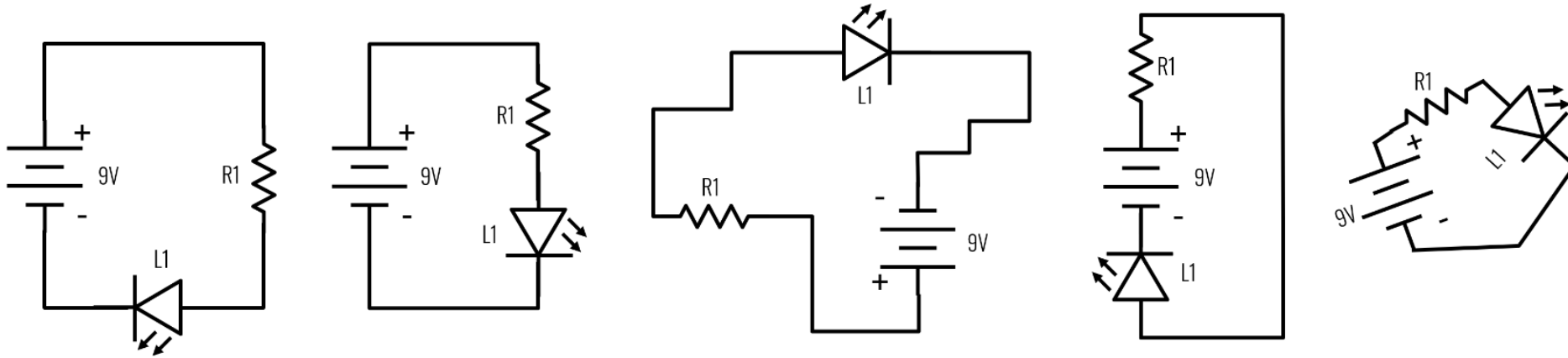
1.10 Circuit schematics (b)

- There are more symbols and models to learn:

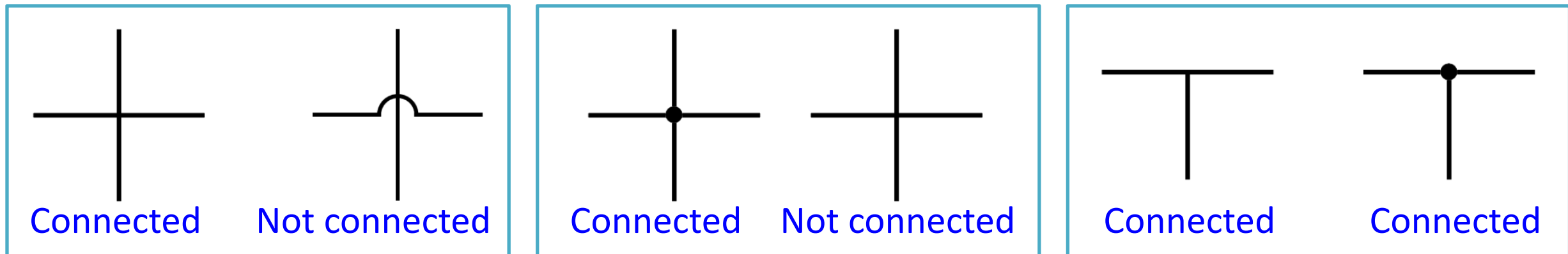


1.10 Circuit schematics (c)

- Connecting by ideal wires: congruent circuit schematics

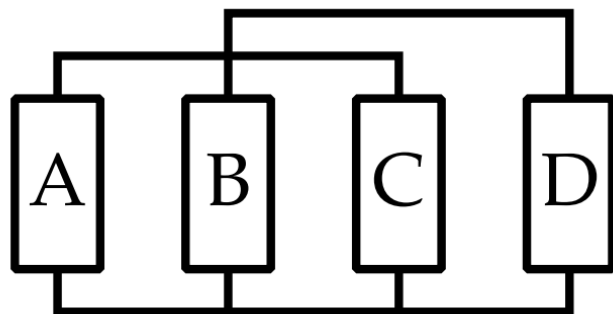
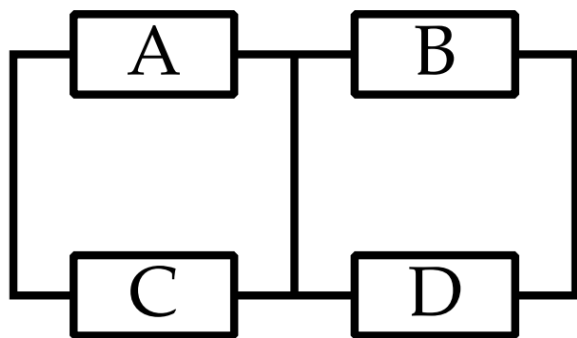


- Wire crossings:



1.10 Circuit schematics (d)

- Congruent circuit schematics and their simplification



Think about what you have learned in your high school.

(1) 电流分支法:

- ①先将各结点用字母标出;
- ②判定各支路元件的电流方向 (若电路原无电压电流, 可假设在总电路两端加上电压后判定);
- ③按电流流向, 自左到右将各元件、结点、分支逐一画出;
- ④将画出的等效图加工整理。

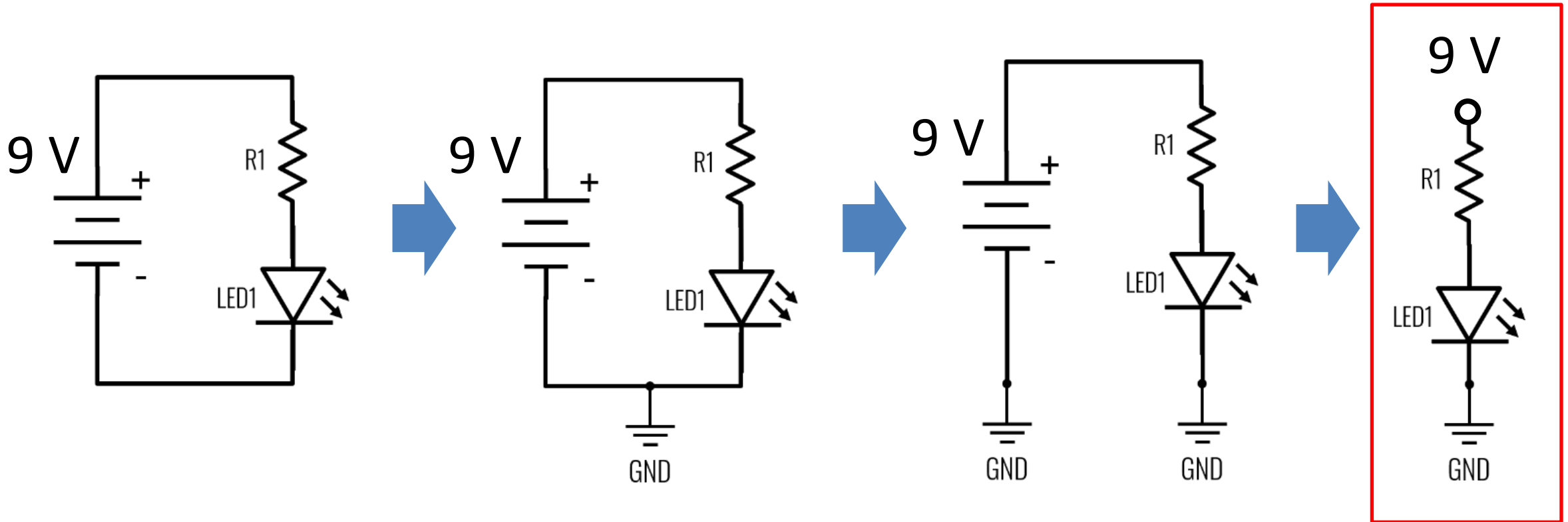
(2) 等势点排列法:

- ①先将各结点用字母标出;
- ②判定各结点电势的高低 (若原电路未加电压, 可先假设加上电压);
- ③将各结点按电势高低自左到右排列, 再将各结点间的支路画出;
- ④将画出的等效图加工整理。

- ①无电流的支路可以去除;
- ②等电势的各点可以合并;
- ③理想导线可以任意长短;
- ④理想电压表可认为断路, 理想电流表可认为短路;

1.10 Circuit schematics (e)

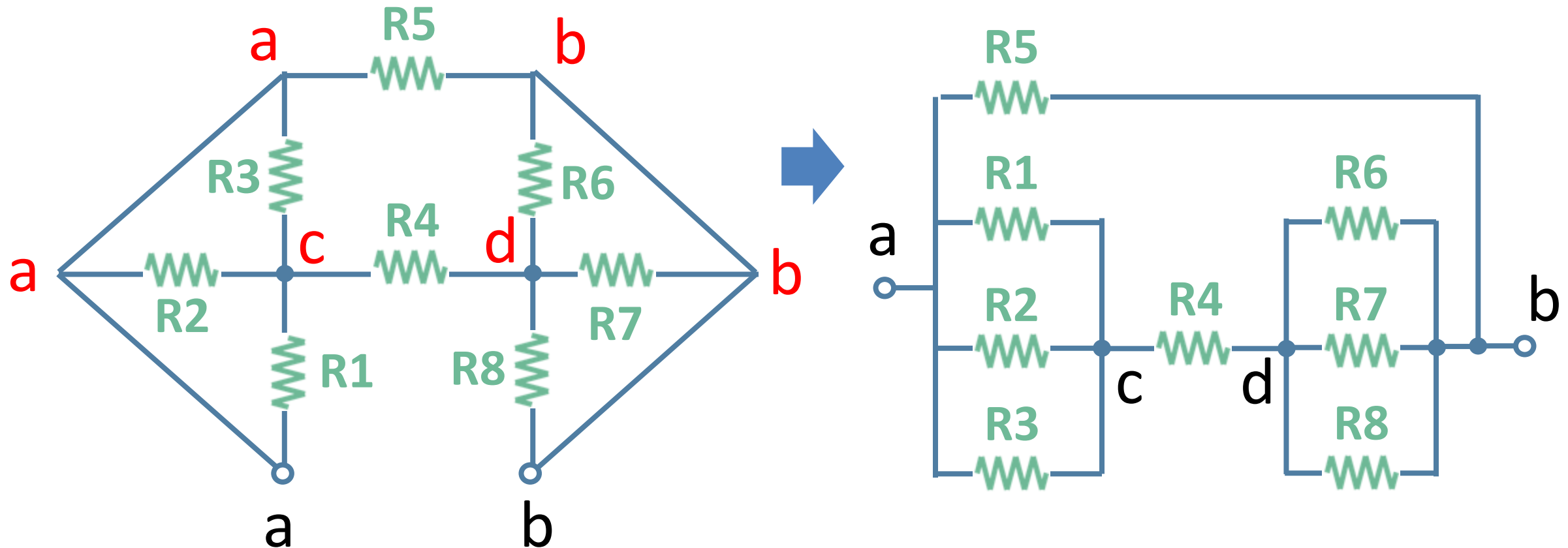
- Different ways for drawing the voltage source and the ground



Four examples of the same circuit drawn differently. The circuit visuals become increasingly minimalist from left-to-right.

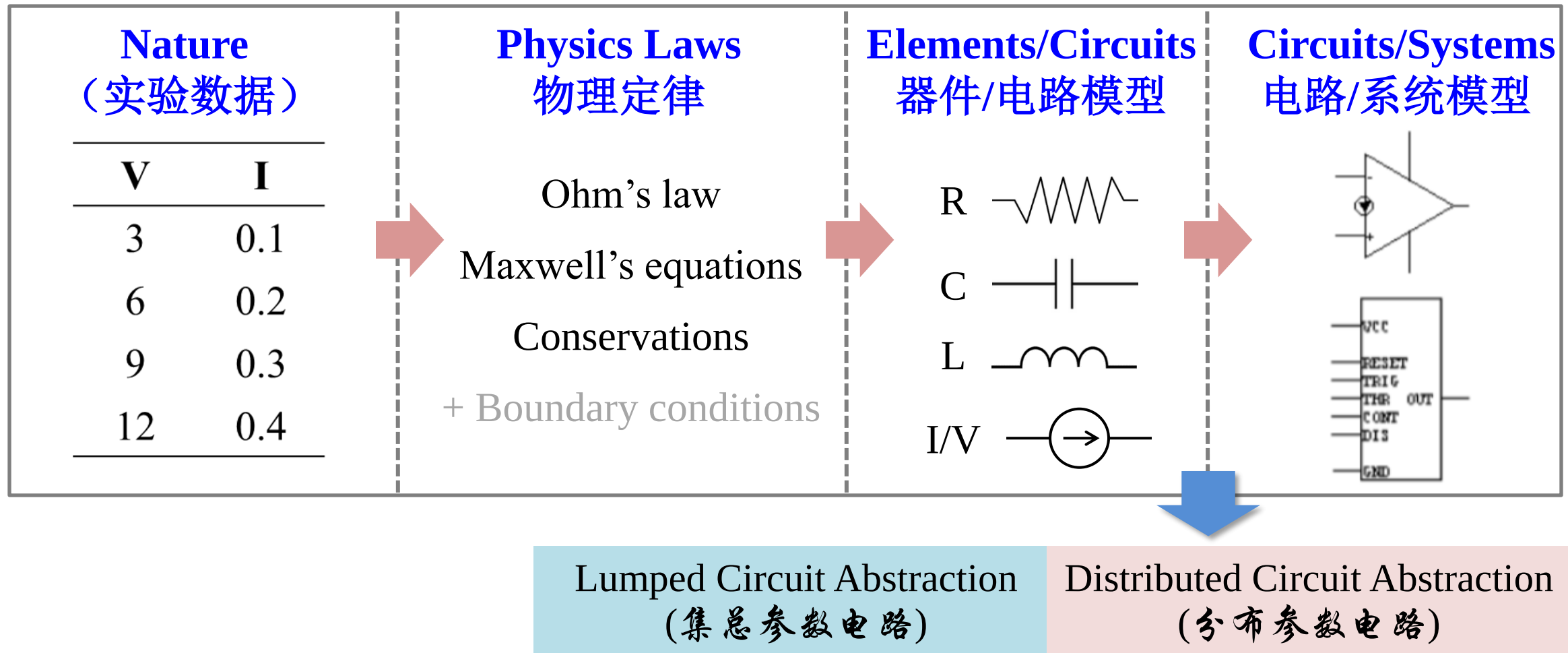
1.10 Circuit schematics (f)

- An example to understand a circuit schematic:



1.11 Lumped Circuit Abstraction (a)

- Proper abstraction/simplification is the art of engineering.



1.11 Lumped Circuit Abstraction (b)

- Lumped Circuit and Distributed Circuit abstractions.

Wave length (λ) V.s. Length of elements (L)

When $\lambda \gg L$

Lumped Circuits (集总参数电路)

- a) V and I **do not vary** across and through conductors connecting elements.
- b) I **does not vary** in two-terminal elements in lumped circuits;
- c) **Neglected transit time** or phase change.

This course, etc.

When $\lambda \approx L$

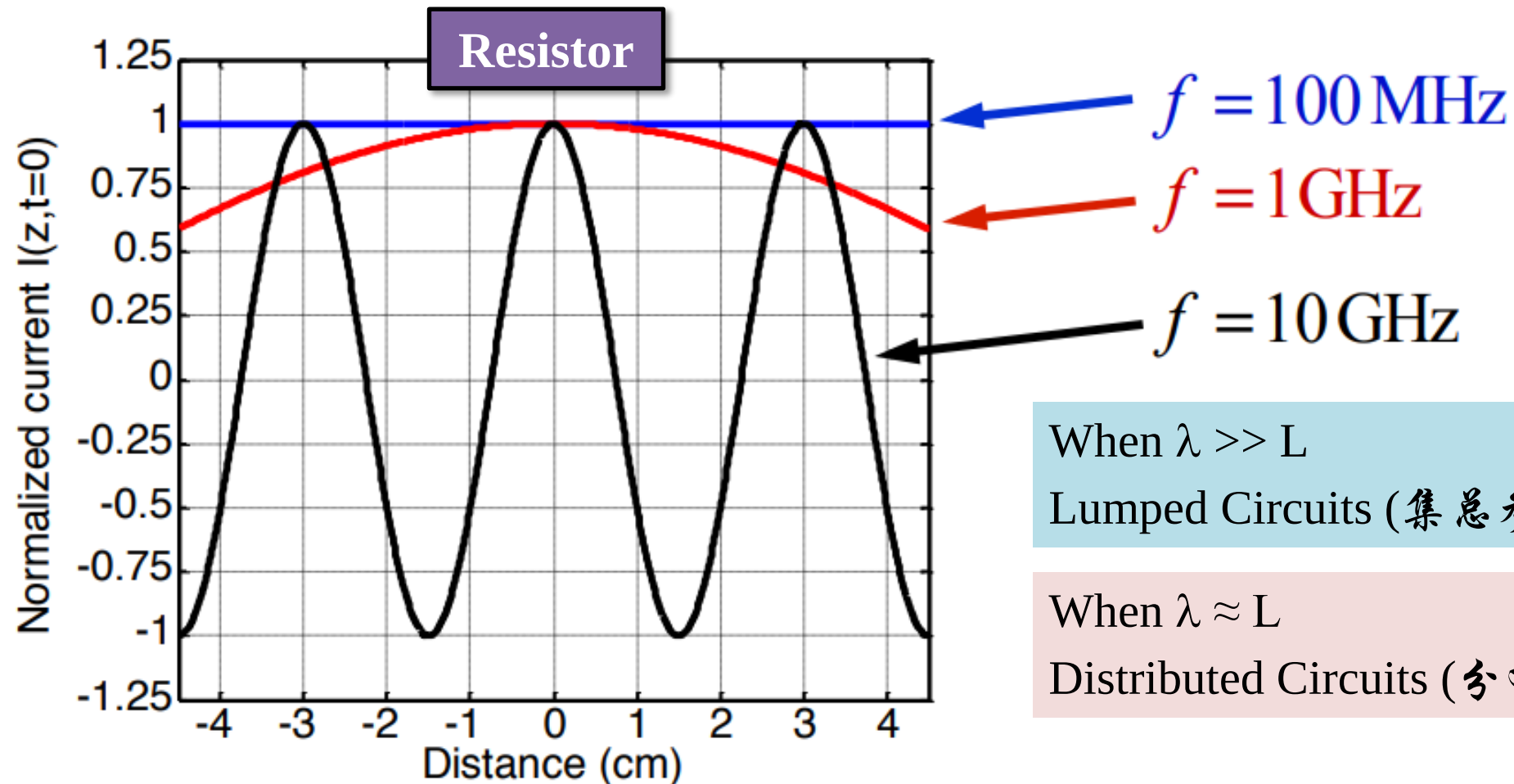
Distributed Circuits (分布参数电路)

- a) I **varies** along conductors and elements;
- b) V **varies** across points along conductor or within element;
- c) **Transit time** and phase change.

EM + RFIC + etc.

1.11 Lumped Circuit Abstraction (c)

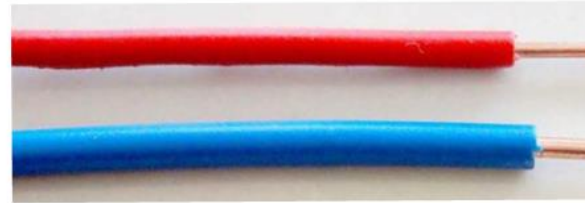
- Frequency versus wavelength.



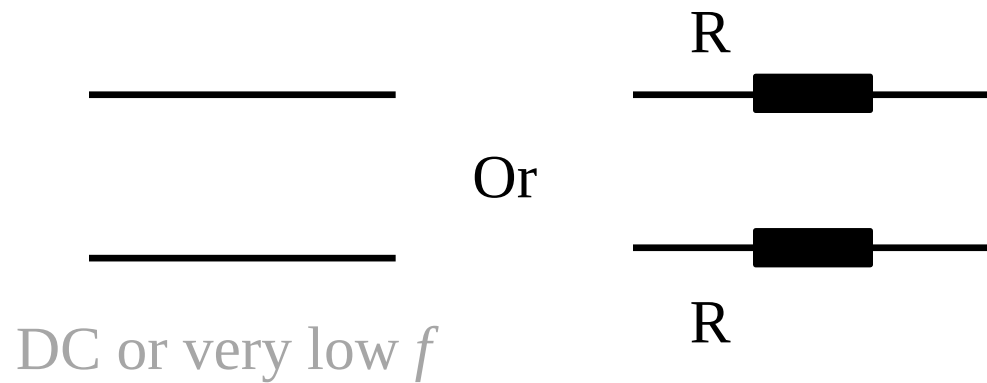
1.11 Lumped Circuit Abstraction (d)

- Models of elements in different cases

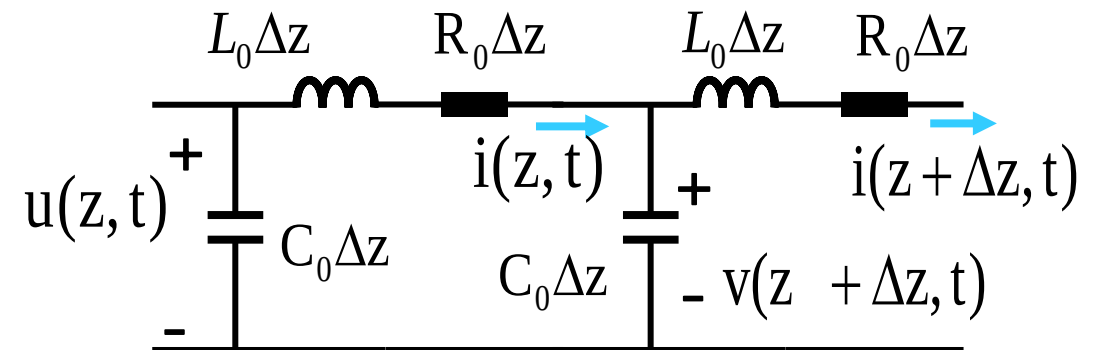
Parallel wires



When $\lambda \gg L$, Lumped Circuits



When $\lambda \approx L$, Distributed Circuits

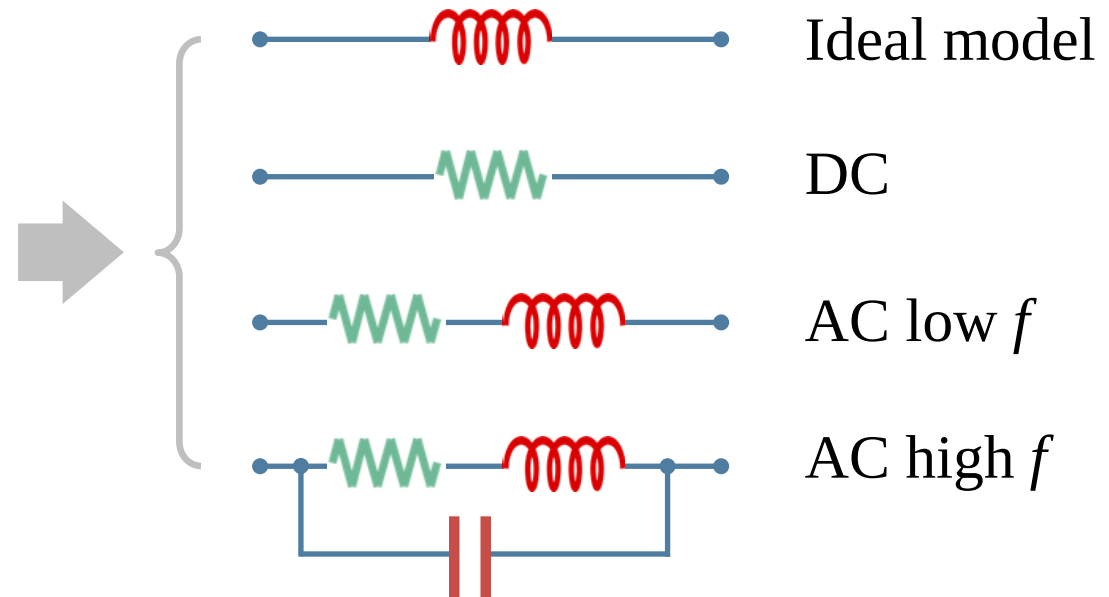
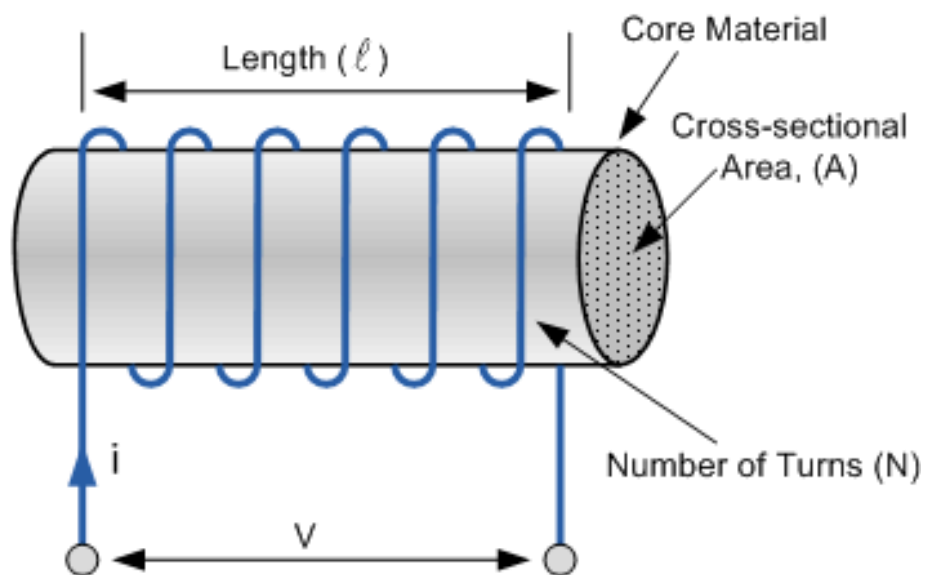


We use different models in different scenarios !!!

1.11 Lumped Circuit Abstraction (e)

- Models of elements in different cases

Inductors



We use different models in different scenarios !!!

1.11 Lumped Circuit Abstraction (f)

- Some examples of distributed circuits.

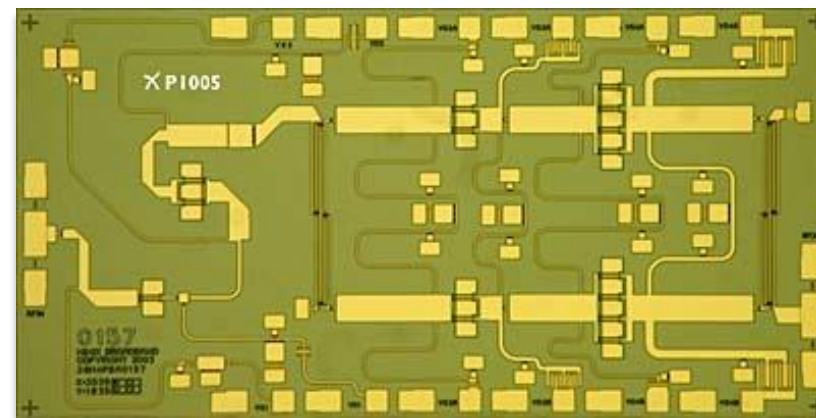


HVAC Transmission (高压交流输电线路)

$$f = 50 \text{ Hz}, \lambda = c/f = 6000 \text{ km}$$

MMIC (微波单片集成电路)

$$f = 5 \text{ Gz}, \lambda = c/f = 6 \text{ cm}$$



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1.11 Summary (a)

- Things you need to know

1. Voltage, current, and power.
2. Reference voltage and current directions.
3. Associated & nonassociated reference directions
4. The law of conservation of energy ($P_{\text{supply}} = P_{\text{absorb}}$)
5. Understand circuit schematics
6. Resistors, independent sources, dependent sources.
7. Lumped circuit abstraction ($\lambda \gg L$)

1. **Electricity is an energy that can be used.**
2. Electric property of particle: **charge.**
3. Collective movement of charge: **current.**
4. Why does a charge move? **Voltage.**
5. Voltage drives the current: **power to do work.**
6. How to measure charge, current, voltage, power? **Unit.**
7. How to use electricity? **Form circuits with elements.**
8. How to design the circuit? **Schematics and models.**
9. How to form the models? **Lumped circuits model.**
10. **Now we are ready to go!**

1.11 Summary (b)

- Important concepts

1. An electric circuit consists of electrical elements connected for a particular purpose.
2. The International System of Units (SI) is the international measurement language, which enables engineers to communicate their results. From the seven principal units, the units of other physical quantities can be derived.
3. Current is the rate of charge flow past a given point in a given direction
4. Voltage is the energy required to move 1 C of charge through an element
5. Power is the energy supplied or absorbed per unit time. It can be calculated as the product of voltage and current.
6. According to the passive sign convention, power assumes a positive sign when the current enters the positive polarity of the voltage across an element.
7. An ideal voltage source produces a specified voltage across its terminals. An ideal current source produces a specified current through its terminals. The specified voltage/current is not affected by what the ideal source is connected to.
8. Voltage and current sources can be dependent or independent. A dependent source is one whose value depends on some other circuit variable.

Cloud Iridescence



2023 春节，珠峰出现彩云（图片来源见水印）